

Bridging the gap between presynaptic hair cell function and neural sound encoding

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Abstract

Neural diversity can expand the encoding capacity of a circuitry. A striking example of diverse structure and function is presented by the afferent synapses between inner hair cells (IHCs) and spiral ganglion neurons (SGNs) in the cochlea. Presynaptic active zones at the pillar IHC side activate at lower IHC potentials than those of the modiolar side that have more presynaptic Ca^{2+} -channels. The postsynaptic SGNs differ in their spontaneous firing rates, sound thresholds and operating ranges. While a causal relationship between synaptic heterogeneity and neural response diversity seems likely, experimental evidence linking synaptic and SGN physiology has remained difficult to obtain. Here, we aimed at bridging this gap by *ex vivo* paired recordings of IHCs and postsynaptic SGN boutons with stimuli and conditions aimed to mimic those of *in vivo* SGN-characterization. Synapses with high spontaneous rate of release (SR) were found predominantly on the pillar side of the IHC. These high SR synapses had larger and more temporally compact spontaneous EPSCs, lower voltage-thresholds, tighter coupling of Ca^{2+} channels and vesicular release sites, shorter response latencies and higher initial release rates. This study indicates that synaptic heterogeneity in IHCs directly contributes to the diversity of spontaneous and sound-evoked firing of SGNs.

Significance Statement

Sound encoding relies on spiral ganglion neurons (SGNs) with diverse spontaneous firing, sound thresholds of firing and sound-intensity range over which SGN firing rate changes. Such functional SGN diversity might originate from different input from afferent synapses with inner hair cells (IHCs). The present study addresses this hypothesis by using recordings from individual IHC-SGN synapses of hearing mice under *ex vivo* conditions aimed to mimic cochlear physiology. The results provide evidence that synaptic heterogeneity in IHCs contributes to SGN firing diversity. Thus, the cochlea employs heterogeneous synapses to decompose sound information into different neural pathways that collectively inform the brain about sound intensity.

eLife Assessment

This **fundamental** study advances substantially our understanding of sound encoding at synapses between single inner hair cells of the mouse cochlea and spiral ganglion neurons. Dual patch-clamp recordings—a technical tour-de-force—and careful data analysis provide **compelling** evidence that the functional heterogeneity of these synapses contributes to the diversity of spontaneous and sound-evoked firing by the neurons. The work will be of broad interest to scientists in the field of auditory neuroscience.

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Introduction

Chemical synapses represent diverse and plastic neural contacts that are adapted to the specific needs of neural computation. Synaptic diversity is expressed across the nervous system, within a given circuit and even within the same neuron (recent reviews in Ref. [Nusser, 2018](#); [Wichmann and Kuner, 2022](#)). Synaptic diversity occurs at various levels: from synapse shape and size, to the ultrastructure of pre- and postsynaptic specializations, to their molecular composition. The auditory system harbors striking examples of synaptic diversity. Glutamatergic ribbon synapses in the cochlea, calyceal synapses in the brainstem, and bouton synapses throughout the central auditory system differ greatly from each other ([Moser et al., 2006](#); [Wichmann and Kuner, 2022](#)). Beyond the diversity across synapses formed by different neurons and at different places (e.g. different regions of the brain or frequency (tonotopic) places of the cochlea ([Johnson et al., 2017](#))), heterogeneity exists even among auditory synapses formed by an individual presynaptic inner hair cell (IHC) with its 5–30 postsynaptic spiral ganglion neurons (SGNs, (reviews in: [Gómez-Casati and Goutman, 2021](#); [Meyer and Moser, 2010](#); [Moser, 2020](#); [Reijntjes and Pyott, 2016](#))). Synaptic heterogeneity has been found at different tonotopic positions of the cochlea and is a candidate mechanism for how the cochlea decomposes acoustic information ([Grant et al., 2010](#); [Meyer et al., 2009](#); [Ohn et al., 2016](#); [Özçete and Moser, 2021](#)). For example, the cochlea might use heterogeneous afferent synapses to break down sound intensity information into complementary spike rate codes of SGNs that have been reported along the tonotopic axis of the cochlea for several species ([Huet et al., 2016](#); [Kiang et al., 1965](#); [Sachs and Abbas, 1974](#); [Taberner and Liberman, 2005](#); [Winter et al., 1990](#)).

Decades of *in vivo* recordings from single SGNs have demonstrated functional diversity of SGNs with comparable frequency tuning, i.e., receiving input from IHCs at a given tonotopic place or potentially even the same IHC. Such functional diversity is present in both spontaneous and sound-evoked firing. The spontaneous firing rate (SR) in the absence of sound varies from less than 1 spikes/s to more than 100 spikes/s ([Barbary, 1991](#); [Evans, 1972](#); [Kiang et al., 1965](#); [Schmiedt, 1989](#); [Taberner and Liberman, 2005](#)). In response to increasing sound pressure, SGNs with high SR show a low sound threshold and a steep rise in the spike rate to increasing sound intensities until the rate saturates. SGNs with low SR have a higher sound thresholds, shallower spike rate rise and saturate at higher sound intensities ([Ohlemiller et al., 1991](#); [Winter et al., 1990](#)). Additionally, SGNs show differences in their excitability ([Crozier and Davis, 2014](#); [Markowitz and Kalluri, 2020](#); [Smith et al., 2015](#)), morphological features ([Liberman, 1980](#); [Merchan-Perez and Liberman, 1996](#); [Tsuji and Liberman, 1997](#)) and heterogeneous molecular profiles ([Li et al., 2020](#); [Petitpré et al., 2020](#), 2018; [Shrestha et al., 2018](#); [Sun et al., 2018](#)). Yet, it has remained challenging to demonstrate a causal link of a candidate mechanism to the physiological SGN diversity.

One common approach has been to capitalize on a pioneering study that employed *in vivo* labelling of physiologically characterized SGNs in cats and proposed that synapses formed by low and high SR SGNs segregate on the basal IHC pole (Lieberman, 1982). High SR SGNs preferentially contacted the pillar side of the IHC (facing pillar cells), while low SR SGNs synapsed on the opposite, modiolar side of the IHC (facing the cochlear modiolus). Interestingly, a segregation has also been found for afferent and efferent synaptic properties, as well as molecular and biophysical SGN properties (Frank et al., 2009; Grant et al., 2010; Hua et al., 2021; Kantardzhieva et al., 2013; Liberman et al., 2011; Markowitz and Kalluri, 2020; Merchan-Perez and Liberman, 1996; Meyer et al., 2009; Neef et al., 2018; Ohn et al., 2016; Özçete and Moser, 2021; Shrestha et al., 2018; Sun et al., 2018; Yin et al., 2014). For instance, type I_b and I_c SGNs preferentially synapse on the modiolar side (Sherrill et al., 2019; Shrestha et al., 2018; Sun et al., 2018) and show low SR (Siebald et al., 2023). Pillar synapses are preferentially formed by type I_a SGNs (Shrestha et al., 2018; Siebald et al., 2023), have smaller IHC active zones (AZs) (Lieberman et al., 2011; Ohn et al., 2016; Reijntjes et al., 2020), and activate at voltages as low as the IHC resting potential (Ohn et al., 2016; Özçete and Moser, 2021). The low voltage of activation of release at pillar synapses shown *ex vivo* could underlie the high SR and low sound threshold of firing of SGNs found *in vivo* but direct demonstration of such a link has been missing.

Here, we aimed to bridge the gap between *ex vivo* presynaptic physiology and *in vivo* SGN neurophysiology. We performed paired IHC and SGN bouton recordings in acutely explanted organs of Corti from hearing mice with stimuli and conditions aimed to mimic *in vivo* SGN-characterization. This approach tightly controls IHC Ca^{2+} influx and records the postsynaptic SGN response to glutamate release at a single afferent synapse. Using the rate of spontaneous excitatory postsynaptic currents (rate of sEPSCs, SR) as a surrogate for SGN SR, we demonstrate that high SR synapses have larger and more compact sEPSCs as well as lower voltage thresholds, shorter latencies of evoked EPSCs, tighter Ca^{2+} channel coupling to vesicle release and higher initial release rates. 90% of these high SR synapses were located on the pillar side of the IHC. Our findings suggest that synaptic heterogeneity accounts for much of the SGN firing diversity at a given tonotopic position.

Results

Simultaneous paired patch-clamp recordings from IHCs and one of the postsynaptic SGN boutons were performed on mice after the onset of hearing (postnatal days (p) 14–20). We performed perforated-patch whole-cell configuration from IHCs, held at their presumed physiological resting potential (−58 mV (Johnson, 2015)), and ruptured-patch whole-cell recordings from one of the postsynaptic SGN boutons (Fig. 1). Due to the technical difficulty of establishing the paired recording, typically only one bouton was recorded per IHC. Recordings were made at body temperature and in artificial perilymph-like solution (Wangemann and Schacht, 1996). To establish the paired recording, we approached boutons facing either the pillar or the modiolar side of the IHC in an effort to elucidate synaptic differences between both sides (Fig. 1A). We nickname the synapse location as “pillar” and “modiolar” based on the DIC-image, but note that efforts to stain and image the recorded boutons by fluorescence microscopy were not routinely successful. In addition, the recorded boutons were classified based on their spontaneous rate of synaptic transmission (Fig. 1B, Fig. 2 and related figure supplements; SR, **Low SR** < 1 sEPSC/s vs **High SR** ≥ 1 sEPSC/s according to (Taberner and Liberman, 2005)). We then performed an *in-depth* biophysical analysis of evoked release (Fig. 1C and Figs. 3–4, and related figure supplements).

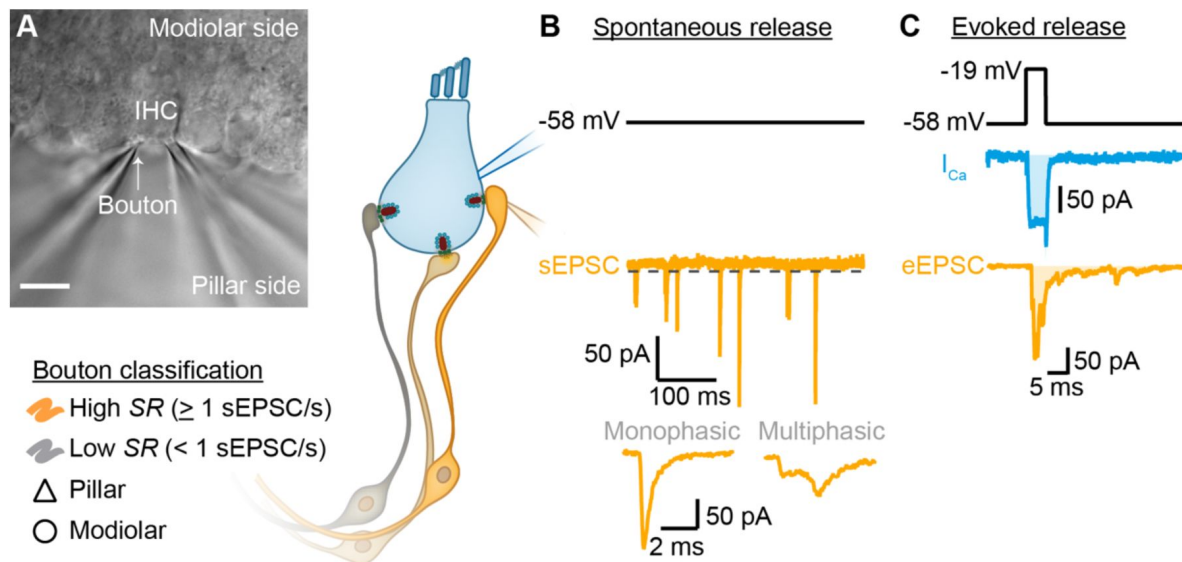


Figure 1.

Paired IHC-bouton patch-clamp recordings to study the release properties of individual IHC ribbon synapses as a function of synapse position

A. Differential Interference Contrast image of an explanted murine organ of Corti. In this example, supporting cells from the pillar side were removed to gain access to the IHCs and their contacting boutons. The recorded boutons were classified based on their position (Δ pillar or $\circ$ modiolar) and on their spontaneous rate (SR) (Low SR < 1 sEPSC/s vs High SR ≥ 1 sEPSC/s). Scale bar: 10 μ m. **B.** Spontaneous release was recorded in absence of stimulation (i.e., IHC holding potential = -58 mV; Supplementary Table 1; dashed line represents the threshold for sEPSC detection). sEPSCs were classified as monophasic (a steady rise to peak and monoexponential decay, temporally more compact) or as multiphasic (multiple inflections and slowed raising and decaying kinetics, non-compact). **C.** Evoked release: depolarizing pulses (black trace) were used to trigger whole IHC Ca^{2+} influx (I_{Ca} , blue trace) and ensuing release of neurotransmitter that evoked EPSCs (eEPSCs, light orange trace). Ca^{2+} charge and eEPSC charge were estimated by taking the integral of the currents (shaded light blue and light orange areas).

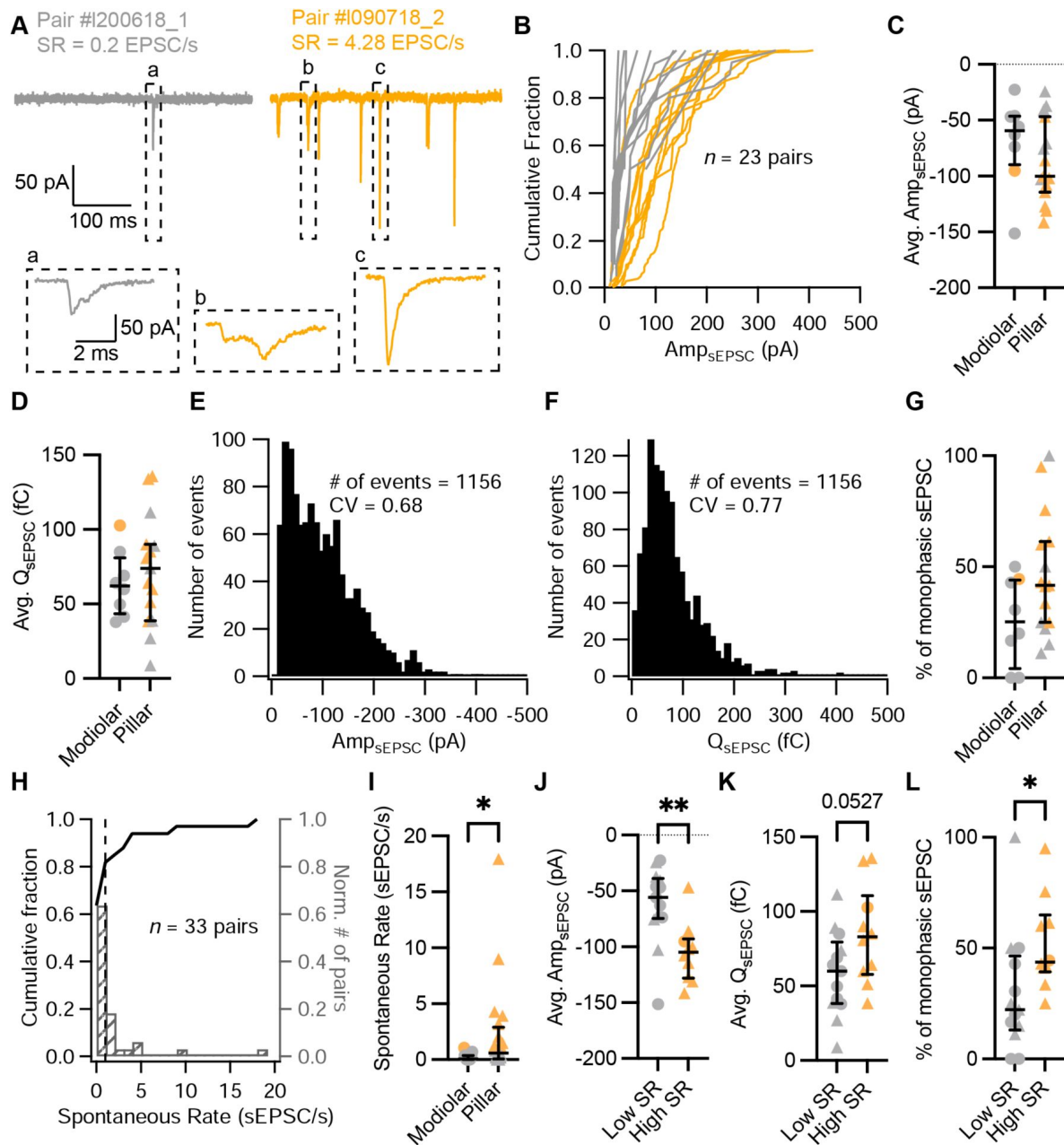


Figure 2.

Synapses with high spontaneous release have larger and more monophasic sEPSCs

A. Spontaneous EPSCs recorded in the absence of stimulation (i.e., IHC holding potential = -58 mV) from two exemplary paired recordings with different spontaneous rate (SR: grey for low SR, orange for high SR). 'Pair #' identifies individual paired recordings. Insets show the selected sEPSCs in an expanded time scale. (a,b) correspond to multiphasic sEPSCs, while (c) represents a typical monophasic sEPSC. **B.** Cumulative sEPSC amplitude plots for 23 paired synapses that had spontaneous release. **C-D.** Average sEPSC amplitude (C) and charge (D) from individual synapses recorded from the pillar or modiolar side of the IHC. **E-F.** Pooled sEPSC amplitude (E) and charge (F) distributions show a distinct peak at -40 pA and 40 pC, respectively. Bin size: 10 pA or pC. **G.** Percentage of monophasic sEPSCs in pillar and modiolar synapses. **H.** Cumulative fraction (left axis) and normalized histogram (right axis) of the spontaneous rate (bin size is 1 sEPSC/s) of 33 pairs. **I.** Pillar synapses had higher rates of sEPSCs. **J-L.** High SR synapses had significantly larger sEPSC amplitudes (J), a tendency to bigger sEPSC charges (K) and higher percentages of monophasic sEPSCs (L).

Panels G, I-L show individual data points with the median and interquartile range overlaid (line). Synapses were classified as $\triangle$ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

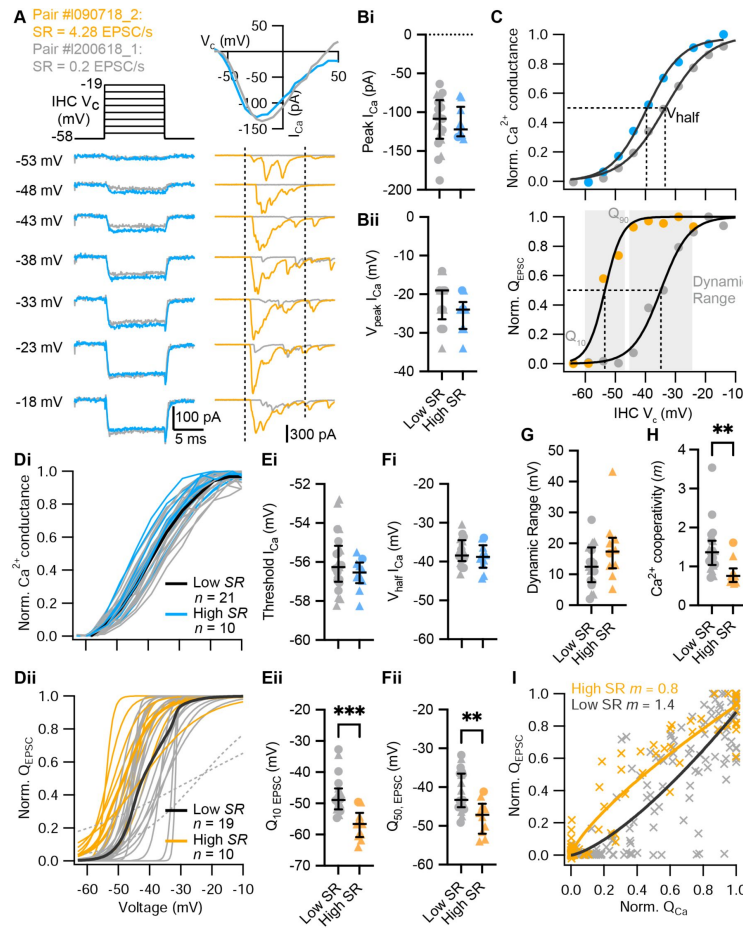


Figure 3.

High SR synapses activate at lower voltages and show tighter Ca^{2+} channel coupling of synaptic release

A. Voltage-protocol (top left), IHC Ca^{2+} current (I_{Ca} , bottom left and top right, blue and gray) and eEPSCs (bottom right, orange and gray) of a high and a low SR pair, respectively, in response to 10 ms depolarizations (dashed vertical lines on top of eEPSC data indicate the onset and offset of the depolarization) to different potentials ranging from -58 to -19 mV in 5 mV steps. The upper right panel shows the current-voltage relationships for the two pairs. **Bi-Bii.** The peak of whole-cell Ca^{2+} current (Bi) and the voltage eliciting maximum Ca^{2+} current (Bii) of IHCs were comparable between high and low SR synapses. **C.** Upper panel: Fractional activation of the Ca^{2+} channels (blue and gray data points from the examples shown in A) was obtained from the normalized chord conductance. Voltage of half-maximal activation ($V_{half} Ca$; dotted line) and voltage sensitivity of activation (slope k) were determined using a Boltzmann fit (black trace) to the activation curve. Lower panel: Release-intensity curve (orange and gray data points from the examples shown in A) was obtained from the Q_{EPSC} for each depolarization step. A sigmoidal function (black trace) was fitted to obtain the voltage of half-maximal synaptic release ($V_{half} Q_{EPSC}$; dotted line) and the voltage sensitivity of the release (slope), as well as the dynamic range for which the exocytosis changes from 10-90% (gray area). **Di-Dii.** Voltage dependence of whole-cell Ca^{2+} channel activation (activation curve; Di) and fits to release-intensity curves (Dii) for 31 synapses. Averages (thick lines) and individual curves (thin lines) are overlaid. The release-intensity curve of two low SR pairs could not be fitted (grey dotted lines). **Ei-Fi.** The threshold of Ca^{2+} influx (Ei) and $V_{half} I_{Ca}$ (Fi) did not differ between low and high SR synapses. **Eii-Fii.** Voltage of 10% maximum release ($Q_{10} EPSC$, Eii) and $V_{half} Q_{EPSC}$ (Fii) were significantly more hyperpolarized in high SR synapses. **G.** Dynamic range of release was comparable between low and high SR synapses. **H.** Ca^{2+} cooperativity (m) estimated from fitting a power function to the $Q_{EPSC} - Q_{Ca}$ relationship for each individual synapse (see **Figure 3-figure supplement 2**) was significantly lower in high SR synapses. **I.** Scatter plot of normalized Q_{EPSC} versus the corresponding normalized Q_{Ca} . The solid lines are a least-squares fit of a power function ($Q_{EPSC} = a(Q_{Ca})^m$) to the data yielding $m_{high SR}$ of 0.8 and $m_{low SR}$ of 1.4. Panels B, E-H show individual data points with the median and interquartile range overlaid (line). Synapses were classified as Δ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

Spontaneous synaptic transmission

In order to recapitulate synaptic transmission in the absence of sound stimulation, we held the IHC at their presumed physiological resting potential (-58 mV (Johnson, 2015)). Since $\text{Ca}_v1.3$ Ca^{2+} -channels activate at low voltages (-65 to -45 mV (Koschak et al., 2001; Picher et al., 2017; Platzner et al., 2000; Xu and Lipscombe, 2001)), their open probability at the IHC resting potential is thought to be sufficient to trigger spontaneous release (Glowatzki and Fuchs, 2002; Özçete and Moser, 2021; Robertson and Paki, 2002). Under our experimental conditions, spontaneous, i.e., excitatory postsynaptic currents in the absence of IHC stimulation (sEPSCs) were observed in 23 of 33 pairs (Fig. 1B, Fig. 2, 20 recordings were targeted to the pillar side and 13 to the modiolar side). We used the rate of sEPSCs (SR) as a surrogate of SGN SR as sEPSCs trigger SGN firing with >90% probability (Rutherford et al., 2012). Regardless of all efforts to maintain the physiological integrity in the *ex vivo* experiments, we expect our estimated rates of sEPSC to underestimate the SGN SR for the same age group (Wong et al., 2013). For each paired recording, we quantified the SR during 5 or 10 s of a continuous recording, or during the segment before and after the step depolarization protocols (Supplementary Table 1). Amplitudes of sEPSCs typically varied from around -10 pA to -400 pA across all recorded IHC-SGN synapses (Fig. 2B). The amplitude histogram for all pairs was slightly skewed towards smaller amplitudes (skewness of 1.06) with a coefficient of variation (CV) of 0.68 (a similar distribution was obtained from 12 bouton-only recordings during which the IHC was not patched [Figure 2-figure supplement 1, panel A]). The charge distribution for all pairs displayed a prominent peak at 40 fC, with a skewness of 2.00 and a CV of 0.77. SR ranged from 0 to about 18 sEPSC/s (Fig. 2H; a similar range of 0 to about 16 sEPSC/s was recorded without patch-clamping the IHC). The SR distribution was highly skewed, with a median of 0.2 sEPSC/s. Following the spontaneous firing rate classification of mouse SGNs by Taberner and Liberman (2005), we classified the synapses into low (< 1 sEPSC/s, ~70%) or high (≥ 1 sEPSC/s, ~30%) SR synapses (see Fig. 2A for examples). Next, we analyzed the recordings as a function of i) synapse position and ii) rate of spontaneous synaptic transmission.

i) Position dependence of synaptic transmission

Of the 33 obtained paired recordings, 20 were classified as pillar synapses. The mean amplitude (Fig. 2C) and charge (Fig. 2D) of the sEPSCs were comparable between modiolar and pillar SGN boutons ($\text{Amp}_{\text{sEPSC}} -69.37 \pm 13.91$ pA [$n = 8$ modiolar] vs -87.21 ± 9.49 pA [$n = 15$ pillar]; $p = 0.2913$, unpaired t-test; $Q_{\text{sEPSC}} 63.74 \pm 7.77$ fC [modiolar] vs 72.54 ± 9.57 fC [pillar]; $p = 0.5463$, unpaired t-test). sEPSC of pillar SGN boutons showed significantly shorter 10-90% rise times than the modiolar ones (Figure 2-figure supplement 1, panel B and C; 0.38 ± 0.04 ms vs 0.57 ± 0.06 ms; $p = 0.0111$, unpaired t-test), yet similar decay times and full-width half-maximum (FWHM, Figure 2-figure supplement 1, panels B, D-E; $p = 0.7997$ and $p = 0.9198$, respectively, unpaired t-test). As a second approach to sEPSC properties, we quantified the percentage of monophasic (or temporally more compact) sEPSCs (Chapochnikov et al., 2014; Glowatzki and Fuchs, 2002) and found a non-significant trend towards higher percentages of monophasic sEPSCs for pillar synapses (Fig. 2G; $25.57 \pm 6.9\%$ for modiolar vs $46.65 \pm 7.06\%$ for pillar; $p = 0.0681$, unpaired t-test). Finally, SR was significantly higher for the pillar synapses (Fig. 2I, mean SR of 2.30 ± 0.96 sEPSC/s; median 0.59; $n = 20$) compared to modiolar ones (Fig. 2I, mean SR of 0.22 ± 0.09 sEPSC/s; median 0.05; $n = 13$; $p = 0.0311$, Mann-Whitney U test).

ii) Relation of synaptic properties to the rate of spontaneous synaptic transmission

High SR synapses had significantly larger sEPSCs (Fig. 2J and Figure 2-figure supplement 1, panel Fi; average sEPSC amplitude of -105.2 ± 8.47 pA for high SR [$n = 10$] vs. -62.39 ± 9.67 pA for low SR [$n = 13$]; $p = 0.0042$, unpaired t-test). sEPSC charge tended to be larger in high SR SGN

synapses (**Fig. 2K** and **Figure 2-figure supplement 1**, panel Fii; $Q_{\text{SEPSC}} 84.23 \pm 10.38$ fC for high SR vs. 58.14 ± 7.79 fC for low SR; $p = 0.0527$, unpaired t-test). The fraction of monophasic sEPSCs was significantly higher in high SR synapses (**Fig. 2L**; $52.08 \pm 6.65\%$; median 43.71%) compared to low SR synapses ($29.49 \pm 7.43\%$; median 22.22%; $p = 0.0185$, Mann-Whitney U test). High SR synapses also showed a significantly faster 10-90% rise times (**Figure 2-figure supplement 1**, panel G; 0.36 ± 0.04 ms) than low SR synapses (0.51 ± 0.05 ms; $p = 0.0420$, unpaired t-test).

Other sEPSCs kinetics, such as decay time constant and FWHM, were not different between low and high SR pairs (**Figure 2-figure supplement 1**, panels H, I; $p = 0.7969$ and $p = 0.9948$, respectively, unpaired t-test). Taken together, these results indicate that high SR synapses are characterized by sEPSCs with larger amplitudes, faster rising times and a more compact waveform, while significant differences of pillar and modiolar synapses were limited to sEPSC rise times. Yet, 9 out of 10 synapses with $SR \geq 1$ sEPSC were located on the pillar side of the IHC.

Evoked synaptic transmission differs between afferent synapses with high and low SR

Next, we compared the physiology of afferent synapses with high and low SR by adapting stimulation protocols routinely employed for *in vivo* characterization of sound encoding by SGNs. We used step depolarizations to emulate physiological receptor potentials given that mature IHCs of the “high frequency” mouse cochlea have graded receptor potentials that primarily represent the rectified envelope of an acoustic stimulus (i.e. the DC component (Russell and Sellick, 1978)).

i) Stimulus intensity encoding at IHC synapses

Sound intensity encoding by SGNs primarily relies on a spike rate code: the average discharge rate increases with the strength of the acoustic stimuli from threshold to saturation of the response. These so-called “rate level functions” are typically analyzed by fitting a sigmoidal function, of which the range of sound pressure level between 10 and 90% of the maximal discharge rate represents the operational or dynamic range (Sachs and Abbas, 1974; Taberner and Liberman, 2005; Winter et al., 1990). To understand stimulus intensity coding at mouse IHC synapses, we measured whole-cell IHC Ca^{2+} currents and the evoked EPSCs (eEPSCs) of SGNs in 31 paired recordings. We stimulated the IHC with 10 ms depolarizations to different potentials ranging up to 57 mV in 5 mV steps (IV protocol; **Fig. 3A**). We deemed it incompatible with a reasonable productivity of the technically challenging, low-throughput paired recordings to combine them with imaging of Ca^{2+} at single AZs. Therefore, this study relies on analysis of the presynaptic Ca^{2+} influx at the level of the whole IHC (i.e. summing over all synapses and a low density of extrasynaptic Ca^{2+} channels, (Frank et al., 2009; Wong et al., 2014)). IHCs with synapses classified as high ($n = 10$) or low SR ($n = 21$) had similar Ca^{2+} current-voltage (IV) relationships: comparable maximal Ca^{2+} currents (**Fig. 3Bi**; $p = 0.6939$, Mann-Whitney U test) elicited at similar potentials (**Fig. 3Bii**; $p = 0.1795$, unpaired t-test) and comparable reversal potentials (**Figure 3-figure supplement 1**, panel A; $p = 0.4034$, unpaired t-test). The fractional activation of Ca^{2+} channels was determined from the normalized chord conductance of the IHC. Fitting a Boltzmann function to these activation curves (**Fig. 3C**, upper panel), we obtained the voltages of half-maximal activation ($V_{\text{half Ca}}$) and the voltage-sensitivity of activation (slope) of the IHC Ca^{2+} channels.

“Release-stimulus intensity” curves, akin of an *ex vivo* representation of the SGN rate-level function, were constructed from the normalized Q_{EPSC} response obtained during the IV protocol (**Figure 3-figure supplement 1**, panel C). The voltage dependence of synaptic vesicle release per active zone was approximated by the fit of a sigmoidal function to the individual release-intensity curves (**Fig. 3C**, lower panel and **Fig. 3Dii**). From these sigmoidal fits, we obtained voltage of 10%- maximal release ($Q_{10, \text{EPSC}}$), voltage of half-maximal release ($Q_{50, \text{EPSC}}$), voltage of 90%-maximal release ($Q_{90, \text{EPSC}}$) and the voltage sensitivity of release (slope). For two low SR

paired recordings (Fig. 3Dii, grey dotted lines), a sigmoidal function did not properly fit the release-intensity curve (assessed by visual inspection) which led us to exclude them from the statistical analysis.

The voltage dependence of activation of whole-cell Ca^{2+} influx was similar between IHCs contacted by high and low SR boutons: threshold of Ca^{2+} influx (Fig. 3Ei; $p = 0.2393$, unpaired t-test), $V_{\text{half Ca}}$ (Fig. 3Fi; $p = 0.3479$, unpaired t-test) and voltage sensitivity of Ca^{2+} influx (Figure 3-figure supplement 1, panel B; $p = 0.3470$, unpaired t-test) did not differ significantly between IHCs contacted by low or high SR synapses. This seems to rule out a potential scenario in which the diverse SGN firing properties would be caused by varying average properties of Ca^{2+} channels among different presynaptic IHCs. $Q_{50, \text{EPSC}}$ (Fig. 3Fii) of high SR synapses (-47.76 ± 1.4 mV; $n = 10$) was 6.76 ± 2.0 mV more negative compared to low SR synapses (-41.00 ± 1.2 mV; $n = 19$; $p = 0.0021$, unpaired t-test). Accordingly, high SR synapses had lower voltage thresholds of release than low SR synapses (Fig. 3Eii, $Q_{10, \text{EPSC}}$ of -56.91 ± 1.5 mV [median -56.64 mV] vs. -47.39 ± 1.4 mV [median -48.89 mV]; $p = 0.0001$, Mann-Whitney U test). The hyperpolarized shift was not significant for $Q_{90, \text{EPSC}}$ (Figure 3-figure supplement 1, panel D; $p = 0.1706$, unpaired t-test). The voltage sensitivity of release, determined by a slope factor, tended to be lower in high SR (4.16 ± 0.75 mV) than in low SR (2.9 ± 0.35 mV) synapses without reaching statistical significance (Figure 3-figure supplement 1, panel E; $p = 0.0940$, unpaired t-test). The dynamic range, defined as the voltage range for which the exocytosis changes from 10-90% ($Q_{90, \text{EPSC}} - Q_{10, \text{EPSC}}$), tended to be larger for high SR synapses without reaching significance (Fig. 3G; 18.30 ± 3.3 mV for high SR synapses vs 12.78 ± 1.5 mV for low SR synapses; $p = 0.0940$, unpaired t-test). The voltage dependence of activation of Ca^{2+} influx and of synaptic vesicle release did not differ significantly when the synapses were grouped based on their topographical location at the IHC ($n = 12$ modiolar synapses vs $n = 17$ pillar synapses; Figure 3-figure supplement 1, panels F-K). However, pillar SR synapses had a tendency to show lower voltage thresholds of release than modiolar synapses (Figure 3-figure supplement 1, panel Gii; $Q_{10, \text{EPSC}}$ of -52.78 ± 1.8 mV [median -53.73 mV] vs. -47.68 ± 1.8 mV [median -49.30 mV]; $p = 0.0725$, Mann-Whitney U test). Altogether, these results demonstrate that high SR synapses release at more hyperpolarized voltages than low SR synapses.

Finally, we studied the apparent Ca^{2+} dependence of synaptic vesicle (SV) release during the aforementioned IV protocol, i.e., in the range of IHC receptor potentials. This protocol varies Ca^{2+} influx mainly via changing the channel open probability and to a lesser extent by changing the single channel current. We note that a supralinear intrinsic Ca^{2+} dependence of exocytosis in IHCs (i.e. Ca^{2+} cooperativity, $m \sim 3-4$ when changing the single channel current) has been observed for IHCs of the cochlear apex in mice after hearing onset (Brandt, 2005; Jaime Tobón and Moser, 2023; Özçete and Moser, 2021; Wong et al., 2014). This is thought to reflect the cooperative binding of ~ 4 Ca^{2+} ions required to trigger IHC exocytosis (Beutner et al., 2001). In contrast, a lower Ca^{2+} cooperativity was observed in these studies when primarily changing the number of open Ca^{2+} channels ($m < 2$). This difference in m observed for the apparent Ca^{2+} dependence of exocytosis has been taken to suggest a tight, Ca^{2+} nanodomain-like control of release sites by one or few Ca^{2+} channel(s) in line with classical studies of the Ca^{2+} dependence of transmitter release (Augustine et al., 1991). Here, we related changes of release at individual synapses (ΔQ_{EPSC}) to the change of the integrated IHC Ca^{2+} influx ($\tau I Q_{\text{Ca}}$). We fitted power functions ($Q_{\text{EPSC}} = a(Q_{\text{Ca}})^m$) to the relationships for individual synapses (Figure 3-figure supplement 2) and found Ca^{2+} cooperativities of $m < 2$ for all but 2 synapses. This result suggests a tight, Ca^{2+} nanodomain-like control of release sites by one or few Ca^{2+} channel(s). Interestingly, however, high SR synapses, on average, had significantly lower Ca^{2+} cooperativities than low SR synapses (Fig. 3H; m_{highSR} of 0.8 ± 0.1 [median 0.75; $n = 10$] vs m_{lowSR} of 1.4 ± 0.1 [median 1.37; $n = 21$]; $p = 0.0016$, Mann-Whitney U test). The fit to pooled normalized data of high and low SR synapses yielded the same Ca^{2+} cooperativities of m_{highSR} of 0.8 and m_{lowSR} of 1.4 (Fig. 3I). When grouped based on their modiolar or pillar location, pillar synapses showed significantly lower Ca^{2+} cooperativities than modiolar synapses (Figure 3-figure supplement 1, panel L; m_{pillar} of 1.0 ± 0.08 [median 0.88; $n =$

19] vs m_{modiolar} of 1.6 ± 0.2 [median 1.3; $n = 12$]; $p = 0.0202$, Mann-Whitney U test). Our findings indicate that most afferent IHC synapses of hearing mice employ a tight, Ca^{2+} nanodomain-like control of release sites by one or few Ca^{2+} channel(s) for physiological sound encoding. Yet, quantitative differences in coupling seem to exist between high SR/pillar synapses and low SR/modiolar synapses, whereby a control of SV release by ~ 1 Ca^{2+} channel prevails at high SR/pillar synapses.

ii) Synaptic vesicle pool dynamics at individual IHC active zones

In 13 of the 31 aforementioned paired recordings (6 classified as low SR and 7 as high SR; 2 belonging to modiolar and 11 to pillar synapses), we employed a forward masking paradigm to study SV pool dynamics of single afferent synapses. The forward masking paradigm (Harris and Dallos, 1979) is commonly used for *in vivo* analysis of SGN spike rate adaptation and recovery from adaptation, which has been attributed to the depletion of readily releasable pool of SVs (RRP) and the recovery from depletion (Avissar et al., 2013; Frank et al., 2010; Furukawa and Matsuura, 1978; Goutman, 2017; Goutman and Glowatzki, 2007; Li et al., 2009; Moser and Beutner, 2000; Schroeder and Hall, 1974). Typically, the *in vivo* protocol is applied at saturating sound pressure levels, which we aimed to mimic using strong step IHC depolarizations (to -19 mV from -58 mV) separated by different interstimulus intervals (ISI: 4, 16, 64 and 256 ms) (Fig. 4A). In analogy to the *in vivo* forward masking paradigm, the first stimulus - called *masker*, as it depresses the response to a subsequent stimulus when applied in rapid succession - had a duration of 100 ms. The second stimulus -denominated *probe*- lasted for 15 ms. The recordings included a time frame of 400 ms preceding the “masker” and 400 ms following the “probe”, and the interval between masker and masker was 20 s. Applied to recordings of eEPSCs, the forward masking protocol provides experimental access to the initial RRP release rates, kinetics and extent of RRP depletion, sustained exocytosis, as well as recovery from RRP depletion. To accommodate the stochasticity of SV release from the RRP of IHC active zones, we run each protocol several times (≥ 3 to ≤ 20), which is routinely done for *in vivo* SGN physiology, but challenging *ex vivo* given the fragile and typically short-lived paired pre- and postsynaptic recordings (e.g. Goutman, 2017; Goutman and Glowatzki, 2007). Note that we did not employ cyclothiazide to inhibit AMPA receptor desensitization and reduce its contribution to postsynaptic eEPSC depression (Goutman, 2017), given the potential presynaptic effects of cyclothiazide in synaptic release (Diamond and Jahr, 1995; Dittman and Regehr, 1998).

For the analysis of evoked release dynamics, we focused on the response evoked by the masker. At the presynaptic level, there was no difference in the peak, initial and final IHC Ca^{2+} currents (I_{Ca}) and Ca^{2+} current charge (Q_{Ca}) between the recordings from high and low SR synapses (Figure 4-figure supplement 1, panel A-D). We calculated the synaptic delay from the onset of the masker to the onset of the eEPSC. High SR synapses had significantly shorter synaptic delays (Fig. 4B), with mean latencies of the first eEPSC after stimulus onset of 1.17 ± 0.09 ms (median 1.18) compared to 3.34 ± 1.04 ms (median 2.09) in low SR pairs ($p = 0.0082$, Mann-Whitney U test). This result was corroborated in a bigger sample size, when we compared the synaptic latencies of the 31 pairs to 10 ms pulses to -19 mV (Figure 4-figure supplement 1, panel E; latency of 1.19 ± 0.14 ms [median 1.14] in high SR compared to 2.57 ± 0.48 ms [median 1.79] in low SR synapses; $p = 0.0101$, Mann-Whitney U test). The latency jitter (measured as the standard deviation of the masker-evoked EPSC latencies; Fig. 4C) was also significantly smaller in high SR synapses compared to low SR synapses (0.38 ± 0.07 ms [median 0.41] vs 1.67 ± 0.72 ms [median 0.98], respectively; $p = 0.0012$, Mann-Whitney U test). Additionally, high SR synapses had significantly larger peak amplitudes of the masker-evoked EPSCs which reports the initial release from the RRP (Figure 4-figure supplement 1, panel F; -420.9 ± 54.22 pA for high SR synapses vs -240.1 ± 23.19 pA for low SR synapses; $p = 0.0149$, unpaired t-test). In contrast, the charge of masker-evoked EPSCs ($Q_{\text{EPSC MASKER}}$) was more comparable (Figure 4-figure supplement 1, panel G; 4.512 ± 0.82 pC for high SR vs 3.020 ± 0.43 pC for low SR pairs; $p = 0.1535$, unpaired t-test). We fitted the first 50 ms of the $Q_{\text{EPSC MASKER}}$ with the sum of a single exponential and a line function in order to analyze RRP depletion and sustained release (Figure 4-figure supplement 1, panel H; dashed

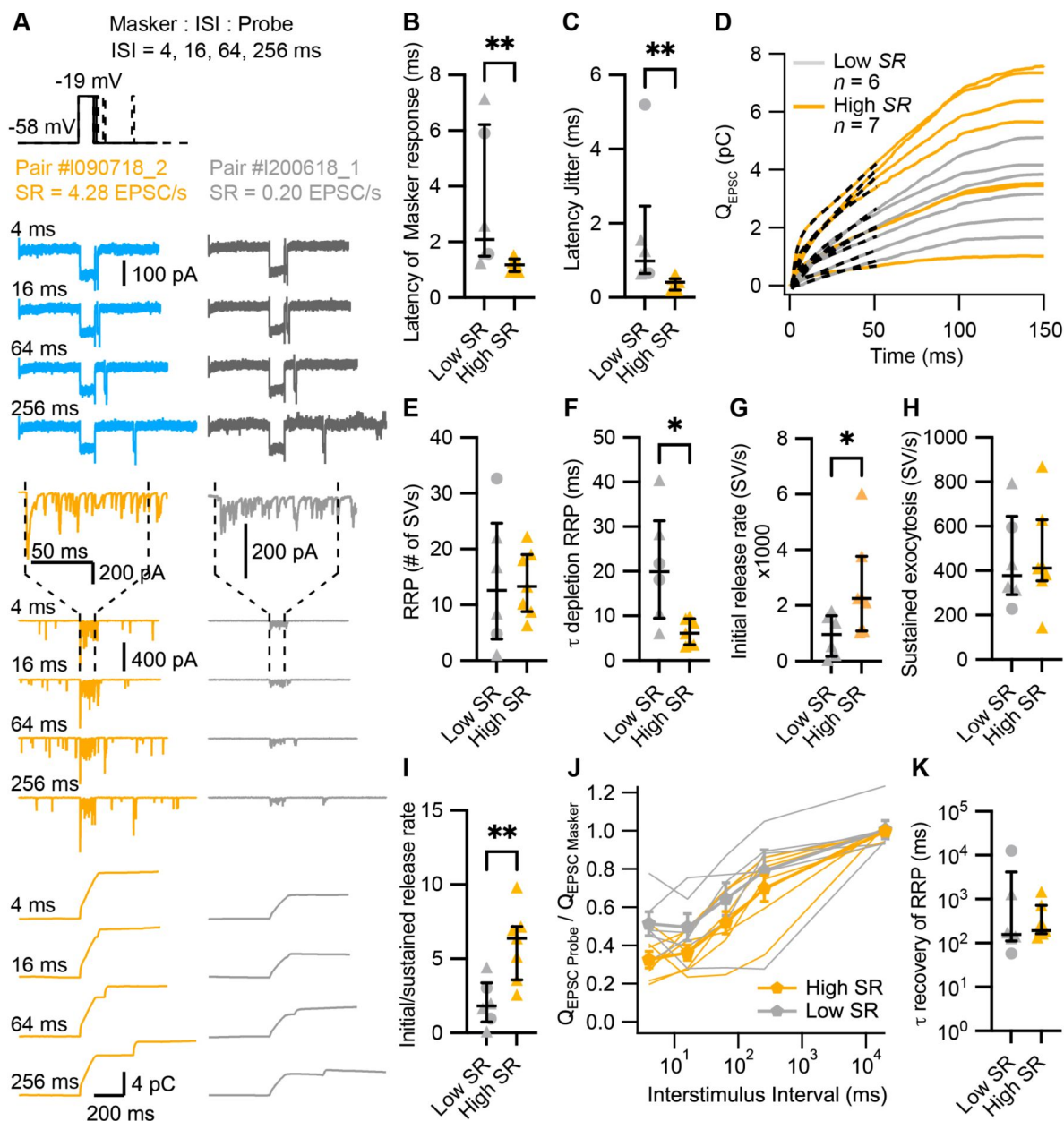


Figure 4.

High SR synapses have shorter synaptic delay and higher initial release rates

A. “Forward Masking” voltage-protocol to study depletion and recovery of RRP and data of an exemplary high SR synapse (left panels, blue and orange) and low SR synapse (right panels, grey): Ca^{2+} currents (2^{nd} from top, I_{Ca}), eEPSCs (2^{nd} from bottom) and Q_{EPSC} (bottom). The stimulus (top panel) consists of two sequential voltage steps (“masker” and “probe”) separated by different interstimulus intervals (ISI). Dashed vertical lines on top of eEPSC data indicate the onset and offset of the masker stimulus. **B.** Latencies of the evoked EPSCs ($\text{EPSC}_{\text{onset}} - \text{Masker}_{\text{onset}}$) were significantly shorter in high SR than low SR synapses. **C.** High SR synapses also had less latency jitter. **D.** Pool depletion dynamics were studied by fitting the sum of a single exponential and a line function (black discontinuous line) to the first 50 ms of the average Q_{EPSC} trace in response to the masker stimulus. **E–I.** RRP, time constant (τ) of depletion, initial release rate and sustained release were calculated from the fits and the mean Q_{EPSC} for each pair. High SR synapses depleted the RRP with faster time constants (F) and reached higher initial release rates (G) followed by a stronger adaptation (I). **J.** Recovery from RRP depletion shown as ratio of Q_{EPSC} probe and Q_{EPSC} masker (mean $\pm$ sem) during the first 10 ms of the stimulus. **K.** Time constant of recovery from RRP depletion obtained from single exponential fits to the traces shown in J (see **Figure 4-figure supplement 1**, panel L).

Panels B, C, E–I and K show individual data points with the median and interquartile range overlaid (line). Synapses were classified as $\triangle$ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

lines in **Fig. 4D**). The amplitude of the exponential component ($A1$), thought to reflect RRP exocytosis, was not different between high and low SR synapses (**Figure 4-figure supplement 1**, panel I; $p = 0.4092$, unpaired t-test). Likewise, the slope of the linear component, reflecting sustained exocytosis, did not differ significantly between the two groups (**Figure 4-figure supplement 1**, panel J; $p = 0.1807$, Mann-Whitney U test).

To quantify synaptic release in terms of SVs, we divided $Q_{\text{EPSC MASKER}}$ by the mean Q_{sEPSC} recorded for each pair (see **Fig. 2K**). This builds on our assumption that each sEPSC corresponds to a unitary release event (“univesicular mode of release” (Chapochnikov et al., 2014; Grabner and Moser, 2018; Huang and Moser, 2018)). The quantal content (RRP size in #SV) was comparable between high and low SR synapses (**Fig. 4E**, 13.99 ± 2.239 SVs vs 14.28 ± 4.839 SVs, respectively; $p = 0.9553$, unpaired t-test). However, the RRP depleted significantly faster in high SR synapses (**Fig. 4F**): $\tau_{\text{RRP depletion}}$ was 6.347 ± 1.096 ms (median 6.13) for high SR vs. 20.88 ± 5.063 ms (median 19.93) for low SR synapses ($p = 0.0140$, Mann-Whitney U test). Accordingly, high SR synapses showed significantly higher initial release rates compared to low SR synapses (**Fig. 4G**; 2651 ± 660.0 SV/s vs. 927.4 ± 307.4 SV/s; $p = 0.0472$, unpaired t-test), which given the comparable RRP size, indicates a higher release probability of the high SR synapses. Moreover, high SR synapses showed stronger depression of the release rate (**Fig. 4I**; the ratio of initial/sustained release rate was 5.941 ± 0.916 for high SR pairs and 2.023 ± 0.6251 for low SR pairs; $p = 0.006$, unpaired t-test), despite comparable sustained release rates (**Fig. 4H**; $p = 0.9258$, unpaired t-test).

Finally, we determined RRP recovery from depletion using the ratio $Q_{\text{EPSC Probe}}/Q_{\text{EPSC Masker}}$ whereby Q_{EPSC} for masker and probe was estimated for the first 10 ms of stimulation. **Figure 4J** plots the ratio for each interstimulus interval (ISI), including the masker-to-masker interval. All the synapses exhibited a smaller response (**Fig. 4J**) and a longer response latency (**Figure 4-figure supplement 1**, panel K) to the probe compared to the response to the masker stimulus. This short-term synaptic depression probably reflects RRP depletion by the masker stimulus (Avisar et al., 2013; Cho et al., 2011; Frank et al., 2010; Moser and Beutner, 2000), in contrast to previous reports of synaptic facilitation when employing short paired stimuli from more hyperpolarized resting potential that do not trigger the full RRP release (Chen and von Gersdorff, 2019; Goutman and Glowatzki, 2011). Surprisingly, and contrary to a previous report in auditory bullfrog synapses (Cho et al., 2011), approximately half of the synapses showed a lower ratio at 16 ms than at 4 ms regardless of their SR. Therefore, to determine the kinetics of recovery from RRP depletion, we fitted a single exponential to the recovery data over the ISI range from 16 ms to 20 s (**Figure 4-figure supplement 1**, panel L). We did not find a significant difference between high SR and low SR synapses (**Fig. 4K**; $\tau_{\text{recovery RRP}}$: 451.5 ± 187.0 ms [median 192.7] vs 2416 ± 2064 ms [median 157.9], respectively; $p = 0.5338$, Mann-Whitney U test). In 4 high SR and 4 low SR synapses, spontaneous activity was resumed shortly after the offset of the probe. The time to this first sEPSC took longer in high SR synapses compared to low ones (**Figure 4-figure supplement 1**, panel M; 185.2 ± 25.25 ms [median 161.6] vs 104.5 ± 24.10 ms [median 108.4], respectively; $p = 0.0286$, Mann-Whitney U test), again indicating stronger synaptic depression on high SR synapses.

Discussion

Much of the information on synaptic sound encoding at afferent IHC-SGN synapses has been obtained from either juxtacellular recordings of SGN firing *in vivo* or from *ex vivo* patch-clamp recordings. Yet, it has remained difficult to reconcile those *in vivo* and *ex vivo* results and to establish a unified account of sound intensity coding in the auditory nerve given differences in experimental conditions, animal models and protocols employed. Here, we biophysically characterized the heterogeneous function of afferent SGN synapses in hearing mice with reference to their rate of spontaneous transmission (SR) as a surrogate of SGN SR that informs their

functional properties. We performed paired pre- and post-synaptic patch-clamp recordings from single IHC synapses of hearing mice under near-physiological conditions using protocols adapted from *in vivo* characterization of SGN's response properties. Using this approach, we were able to distinguish synapses with low and high SR, which we propose to provide the input into low and high SR SGNs. We found that about 90% of high SR synapses were located at the pillar side of the IHC. High SR synapses had larger sEPSCs with a monophasic (or more compact) waveform, lower voltage-thresholds of release, shorter synaptic delays, tighter coupling of release sites to Ca^{2+} channels as well as higher initial release rates and shorter RRP depletion time constants. RRP size, rate of sustained exocytosis and kinetics of RRP recovery from depletion were comparable between high and low SR synapses. We conclude that high SR synapses exhibit higher release probability which likely reflects the tighter coupling of Ca^{2+} channels and release sites. This diversity in the response properties of individual synapses most likely expands the capacity of a single IHC to encode sound intensity over the wide range of audible sound pressures.

Diversity of spontaneous release and their topographical segregation

The SR range observed in our paired recordings from mouse afferent synapses (0 to 18 EPSC/s) agrees with results obtained without patch-clamping the IHC (0 to 16 sEPSC/s) and with previous *ex vivo* reports using loose patch recordings from SGNs of p15-p17 rats (0.1 to 16.42 spikes/s; [Wu et al., 2016](#)). However, the maximum rate is considerably smaller than those recorded *in vivo* from single SGNs of p14-p15 mice (up to 60 spikes/s; [Wong et al., 2013](#)). This 3-fold difference between *ex vivo* and *in vivo* recordings of the same age group could indicate that *in vivo*, the IHC resting potential might be more depolarized and/or subject to spontaneous fluctuations that can trigger Ca^{2+} channel openings and release. Additionally, the presence of K^+ channel blockers (TEA and Cs^+) and differences in pH can all have an impact on the excitability of the cell and the kinetics of the cellular processes.

Other important factors might explain the higher SR reported *in vivo*: i) The intrinsic biophysical properties of SGNs which could further expand the firing rate distribution ([Markowitz and Kalluri, 2020](#)), for instance, due to spikes initiated intrinsically and not activated by an EPSP ([Wu et al., 2016](#)); ii) A sampling bias of synapses with SR above 20 sEPSC/s given that they constitute less than 35% of SGNs from 8-17 weeks old mice ([Taberner and Liberman, 2005](#)); and, iii) the developmental recruitment of high SR SGNs with age ([Niwa et al., 2021](#); [Romand, 1984](#); [Walsh and McGee, 1987](#); [Wong et al., 2013](#); [Wu et al., 2016](#)). 90% of the paired recordings (and 60% for the bouton recordings) of our dataset were obtained from mice between p14-p17, in which spontaneous activity is still low compared to older age groups (p19-p21: 0 – 44.22 spikes/s; p29- p32: 0.11 – 54.9 spikes/s ([Wu et al., 2016](#)); p28: 0 – 47.94 spikes/s, ([Siebald et al., 2023](#))).

In analogy to the pioneering finding of a synaptic segregation of cat SGNs according to SR along the pillar–modiolar axis of IHCs ([Liberman, 1982](#); [Merchan-Perez and Liberman, 1996](#)), we found that about 90% of high SR were located on the pillar IHC side. Yet, not all the synapses of the pillar IHC side had high SR, which agrees with a recent study of molecularly tagged SGNs ([Siebald et al., 2023](#)). These findings suggest that high frequency and large amplitude sEPSCs occur predominantly in synapses with smaller ribbons and AZs, opposing results from retinal cells in which smaller ribbons resulted in reduced frequency and amplitude of EPSCs ([Mehta et al., 2013](#)). It is important to point out that our modiolar/pillar classification is less precise than that of other studies in which the synapse position was quantitatively assigned ([Frank et al., 2009](#); [Liberman et al., 2011](#); [Ohn et al., 2016](#); [Özçete and Moser, 2021](#)). Moreover, other studies also support an overall pillar-modiolar gradient with “salt and pepper” intermingling of synaptic properties rather than their strict segregation ([Ohn et al., 2016](#); [Özçete and Moser, 2021](#)).

Similar to previous reports (e.g. Chapochnikov et al., 2014; e.g. Glowatzki and Fuchs, 2002; Grant et al., 2010; Huang and Moser, 2018; Rutherford et al., 2012), we found a high variability in the waveform and amplitude of sEPSCs between synapses, which apparently do not strictly depend on the topographical location of the synapse (the present work and (Niwa et al., 2021)). Yet, high SR synapses had larger and monophasic (or temporally more compact) sEPSCs. This likely also explains the shorter 10-90% rise time of the sEPSC in high SR synapses, as monophasic sEPSC have shorter rise times than the multiphasic ones (Chapochnikov et al., 2014; Glowatzki and Fuchs, 2002; Huang and Moser, 2018). The difference in the percentage of monophasic sEPSCs and rise times in low and high SR synapses could arise from variability in the fusion pore dynamics on the way to SV fusion and/or on the number of SVs released in timely manner. In the view of the multivesicular hypothesis of spontaneous release (Glowatzki and Fuchs, 2002; Li et al., 2009; Niwa et al., 2021; Schnee et al., 2013), SVs of a modiolar AZ might fuse in an uncoordinated manner, creating EPSCs with a less compact waveform and slower rise times.

Alternatively and our favorite hypothesis, each sEPSC corresponds to a unitary release event (“univesicular mode of release” (Chapochnikov et al., 2014; Grabner and Moser, 2018; Huang and Moser, 2018; Young et al., 2021)) that has the capacity to drive action potential firing (Rutherford et al., 2012). In the framework of the univesicular hypothesis of spontaneous release, the flickering of the fusion pore, prior to or instead of full collapse fusion of the SV, might be favored in low SR synapses, leading to slower sEPSC rise times and a lower percentage of monophasic sEPSCs. Such heterogeneity of fusion pore dynamics has been reported in chromaffin cells, calyx of Held and hippocampal neurons (Chang et al., 2021; Henkel et al., 2019; Shin et al., 2020, 2018).

Finally, we note that sEPSC amplitudes of IHC synapses in hearing mice (present study and (Niwa et al., 2021)) seem lower than in previous *ex vivo* studies on IHC synapses of hearing rats (Chapochnikov et al., 2014; Grant et al., 2010; Huang and Moser, 2018; Young et al., 2021). In rats, the EPSC amplitude distribution changes with maturation, from highly skewed to the left with a peak around -30 pA to a Gaussian-like distribution with a peak at -375 pA (Grant et al., 2010). This does not seem to be the case in mouse IHC synapses. Average EPSC amplitudes in pre-hearing mice are around -100 to -150 pA (Chapochnikov et al., 2014), even with 40 mM K⁺ stimulation (Jing et al., 2013; Takago et al., 2019). On the contrary, mean EPSC amplitudes in hearing mice remained small (around -100 pA) in resting conditions ((Niwa et al., 2021) and the present study), but became significantly larger upon stimulation with 40 mM K⁺ (Niwa et al., 2021) or voltage depolarizations (the present study).

Candidate mechanisms distinguishing evoked release at low and high SR synapses

The temporal and quantal resolution offered by paired recordings allowed us to analyze the biophysical properties of evoked synaptic transmission in relation to the SR of the given synapse. In an intriguing resemblance with *in vivo* evoked firing properties of high SR SGNs (Bourien et al., 2014; Buran et al., 2010; Relkin and Doucet, 1991; Rhode and Smith, 1985; Taberner and Liberman, 2005), high SR synapses showed lower voltage (~sound pressure *in vivo*) thresholds of synaptic transmission (~firing *in vivo*), shorter and less variable synaptic latencies (~first spike latencies *in vivo*), and higher initial release rates (~onset firing rate *in vivo*). In addition, we found stronger synaptic depression at high SR synapses, which agrees well with the finding of a greater ratio of peak to adapted firing rate in high SR SGNs recorded *in vivo* (Taberner and Liberman, 2005). These results support the hypothesis that IHC synaptic heterogeneity (Frank et al., 2009; Hua et al., 2021; Ohn et al., 2016; Özçete and Moser, 2021; Reijntjes et al., 2020) contributes to the diversity of spontaneous and sound-evoked SGN firing.

How do high SR synapses with likely smaller ribbons and lower maximal Ca^{2+} influx achieve a shorter latency and higher initial release rate? Our hypothesis is that, in high SR synapses, a more hyperpolarized Ca^{2+} channel activation (Ohn et al., 2016) in combination with tighter coupling between the Ca^{2+} channels and the Ca^{2+} sensor of fusion (this work and (Özçete and Moser, 2021)) would enable a faster response with a greater initial SV release probability for a given stimulus. The spatial coupling of the Ca^{2+} channel to the SV release site has also been shown to greatly affect release probability in other synapses (Eggermann et al., 2012; Fekete et al., 2019; Moser et al., 2019; Rebola et al., 2019). Thus, a heterogeneous Ca^{2+} coupling would diversify the response properties (i.e., synaptic vesicle release probability) of individual synapses to the same stimulus. In IHCs, this is particularly important for sound intensity and temporal coding (reviewed in (Moser et al., 2023)). Interestingly, genetic disruptions that shifted the voltage-dependence had a greater impact on the *in vivo* distribution of SR and onset firing rate of SGNs than mutations that changed the maximal synaptic Ca^{2+} influx (Jean et al., 2018; Ohn et al., 2016).

Paired pre- and postsynaptic patch-clamp recordings (this work) and single synapse imaging of presynaptic Ca^{2+} signals and glutamate release (Özçete and Moser, 2021) jointly found a lower apparent Ca^{2+} cooperativity in pillar synapses during depolarizations within the range of receptor potentials. The sensitivity and temporal resolution of paired recordings further allowed us to classify the synapses based on their spontaneous rate and support the hypothesis that high SR SGNs receive input from AZ with tighter coupling than low SR SGNs. However, single synapse imaging (Özçete and Moser, 2021) found a wider range of apparent Ca^{2+} cooperativities than our two non-overlapping data sets for paired patch-clamp recordings (this work and (Jaime Tobón and Moser, 2023)). This might reflect two important technical differences: i) single synapse imaging assessed the presynaptic Ca^{2+} influx of the specific synapse, while in paired recordings we related release to the whole cell Ca^{2+} influx, and ii) the temporal resolution of paired recordings allowed to study the initial release rate using shorter stimuli than in imaging, which avoids an impact of RRP depletion and ongoing SV replenishment. Future studies, potentially combining paired patch-clamp recordings with imaging of presynaptic Ca^{2+} signals, will be needed to further scrutinize the heterogeneity of Ca^{2+} dependence of release in IHCs and its impact on release probability.

Other factors that affect release probability include variations in the number of open Ca^{2+} channels at the AZ (Gratz et al., 2019; Holderith et al., 2012; Scimemi and Diamond, 2012; Sheng et al., 2012; Wong et al., 2013) and the fusion competence of the SV (Klenchin and Martin, 2000), including the priming and docking state (Lin et al., 2022; Neher and Brose, 2018). The ensuing Ca^{2+} influx in pillar synapses might facilitate Ca^{2+} channels and priming of SVs (Cho and von Gersdorff, 2012; Goutman and Glowatzki, 2011, 2007; Michalski et al., 2017; Moser and Beutner, 2000; Pangrsic et al., 2015; Schnee et al., 2011; Spassova et al., 2004) and contribute to the observed results in high SR synapses. Regarding the fusion competence of SVs, it is unknown whether modiolar and pillar synapses exhibit different numbers of docked and primed SVs. To date, ultrastructural studies that resolve docked and tethered SVs have not addressed the topographical location of the AZ in the murine IHC (Chakrabarti et al., 2022, 2018).

Besides SV release probability, RRP size co-determines neurotransmitter release. Our estimated RRP of about 14 SVs in both high and low SR synapses compares well to prior estimates obtained using *ex vivo* electrophysiology (10-40 SVs: Refs. (Goutman and Glowatzki, 2007; Jean et al., 2018; Johnson et al., 2005; Khimich et al., 2005; Moser and Beutner, 2000; Pangrsic et al., 2010; Schnee et al., 2005)), model-based analysis of SGN firing (4-40 SVs (Frank et al., 2010; Jean et al., 2018; Peterson et al., 2014)) and electron microscopy (10-16 SVs within 50 nm of the presynaptic membrane (Chakrabarti et al., 2018; Frank et al., 2010; Graydon et al., 2011; Kantardzhieva et al., 2013; Khimich et al., 2005; Pangrsic et al., 2010)). However, previous reports based on electron microscopy (Kantardzhieva et al., 2013; Merchan-Perez and Liberman,

1996 [\[10\]](#); Michanski et al., 2019 [\[11\]](#)) suggested larger pools of SVs at modiolar synapses, while our electrophysiological estimate of RRP size was comparable between low and high SR synapses. This finding argues against a strong contribution of RRP size to the observed differences in neurotransmitter release. However, higher release probability with comparable RRP size explains higher initial release rates, which likely explain the faster and temporally more precise postsynaptic depolarization that is likely to turn into shorter first spike latencies and lower first spike latency jitter (this study and (Buran 2010).

Finally, the heterogeneity in the functional properties of IHC synapses could arise from molecular heterogeneity of the AZ. In central glutamatergic synapses, molecular heterogeneity of synaptic proteins plays a critical role in the modulation of SV release probability and priming state (Neher and Brose, 2018 [\[12\]](#); Wichmann and Kuner, 2022 [\[13\]](#)). For instance, differential isoforms of priming factors and scaffold proteins have been suggested to tune the functional synaptic diversity of central synapses (Fulterer et al., 2018 [\[14\]](#); Rebola et al., 2019 [\[15\]](#); Rosenmund et al., 2002 [\[16\]](#)). Cochlear IHCs have an unconventional fusion machinery that appears to work without neuronal SNAREs (Nouvian et al., 2011 [\[17\]](#)) (but see Calvet et al., 2022 [\[18\]](#)) and priming factors such as Munc13 and CAPS (Vogl et al., 2015 [\[19\]](#)). Therefore, future studies will need to determine the molecular nanoanatomy underlying the specific AZ nanophysiology and functional synaptic heterogeneity at IHCs. Promising candidates include RBPs (Butola et al., 2021 [\[20\]](#); Grauel et al., 2016 [\[21\]](#); Krinner et al., 2021 [\[22\]](#), 2017 [\[23\]](#); Petzoldt et al., 2020 [\[24\]](#)), RIMs (Jung et al., 2015 [\[25\]](#); Maria M. Picher et al., 2017 [\[26\]](#)), and Septin (Fekete et al., 2019 [\[27\]](#); Yang et al., 2010 [\[28\]](#)).

Challenges for relating synaptic and neural response properties

Next to providing support for the presynaptic hypothesis of functional SGN diversity, the present study also highlights some of the challenges met when aiming to bridge the gap between presynaptic hair cell function and neural sound encoding. Despite major efforts undertaken to match experimental conditions and protocols, it remains difficult to reconcile some findings of *ex vivo* and *in vivo* physiology. Parameters such as RRP size (~# spikes of the rapidly adapting component of firing), sustained exocytosis (~adapted firing rate *in vivo*), recovery of spontaneous and evoked release (~recovery from forward masking *in vivo*) did not differ among our high and low SR synapses, and contrasts with *in vivo* data (e.g. Refs. (Bourien et al., 2014 [\[29\]](#); Buran et al., 2010 [\[30\]](#); Relkin and Doucet, 1991 [\[31\]](#); Rhode and Smith, 1985 [\[32\]](#); Taberner and Liberman, 2005 [\[33\]](#))).

Of particular interest is that the dynamic ranges and slope of release-intensity relationship of high and low SR synapses diverge from the expectations if assuming that high SR SGNs are driven by high SR synapses. High SR synapses tended to show broader dynamic ranges with shallower slopes, while, *in vivo*, high SR SGNs show smaller dynamic ranges and steeper slopes than the low SR ones (Ohlemiller et al., 1991 [\[34\]](#); Winter et al., 1990 [\[35\]](#)). Could this reflect the non-linear saturating properties of the basilar membrane (Sachs et al., 1989 [\[36\]](#); Sachs and Abbas, 1974 [\[37\]](#); Yates et al., 1990 [\[38\]](#)) (discussed in Ref. (Ohlemiller et al., 1991 [\[34\]](#))) which might widen the rate level function of low SR SGNs? Or is it due to a partial depletion of the “standing” RRP (i.e. the occupancy of the RRP release sites with a fusion-competent SV (Moser, 2020 [\[39\]](#); Pangrsic et al., 2010 [\[40\]](#))) at high SR synapses *in vivo*? It remains to be determined whether this and the other aforementioned differences between our data and *in vivo* reports could be attributed to mechanisms downstream of glutamate release and AMPA receptor activation. The possible mechanisms include but are not limited to: i) different spike rates due to diverse EPSC waveforms (Rutherford et al., 2012 [\[41\]](#)); ii) differences in SGN excitability (Crozier and Davis, 2014 [\[42\]](#); Markowitz and Kalluri, 2020 [\[43\]](#); Smith et al., 2015 [\[44\]](#)) due to heterogenous molecular (Petitpré et al., 2020 [\[45\]](#), 2018 [\[46\]](#); Shrestha et al., 2018 [\[47\]](#); Sun et al., 2018 [\[48\]](#)) and morphological profiles (Liberman, 1980 [\[49\]](#); Merchan-Perez and Liberman, 1996 [\[50\]](#); Tsuji and Liberman, 1997 [\[51\]](#)); and iii) differences in efferent innervation of SGNs (Hua et al., 2021 [\[52\]](#); Liberman, 1990 [\[53\]](#); Ruel et al., 2001 [\[54\]](#); Wu et al., 2020 [\[55\]](#); Yin et al., 2014 [\[56\]](#)). Certainly, caution is to be applied for the comparison of *ex vivo* and *in vivo* data

due to the partial disruption of the physiological milieu despite our efforts to maintain near-physiological conditions, and the incomplete synaptic maturation when focusing *ex vivo* experiments on the 3rd postnatal week soon after hearing onset.

Clearly more work is needed to elucidate the mechanisms of SGN firing diversity in the cochlea. Ideally, future studies will combine *in vivo* and *ex vivo* experiments, such as combining physiological SGN characterization with neural backtracing and synaptic morphology of labelled SGNs using volume imaging of afferent and efferent connectivity (Hua et al., 2021 [↗](#)). Moreover, combining optogenetic IHC stimulation with imaging of SGN activity could provide higher throughput and serve posthoc morphology. Finally, paired patch-clamp recordings, as done in the present study, could be combined with SGN subtype-specific molecular labeling, fiber tracing and immunolabeling to further relate synaptic transmission and SGN neurophysiology.

Materials and Methods

Animals and tissue preparation

c57BL/6N mice of either sex between postnatal day 14-27 were used. For paired recordings, the number of animals per age was: p14 (9), p15 (7), p16 (9), p17 (5), p18 (2), p20 (1). For the bouton recordings of **Figure 2-figure supplement 1** [↗](#), panel A, the number of animals per age was: p14 (2), p15 (3), p16 (3), p21 (1), p24 (1), p25 (1), p27 (1). The animal handling and experiments complied with national animal care guidelines and were approved by the University of Göttingen Board for animal welfare and the Animal Welfare Office of the State of Lower Saxony. Animals were sacrificed by decapitation and the cochleae were extracted in modified Hepes Hank's solution containing: 5.36 mM KCl, 141.7 mM NaCl, 1 mM MgCl₂·6H₂O, 0.5 mM MgSO₄·7H₂O, 10 mM HEPES, 0.5 mg/ml L-glutamine, and 1 mg/ml D-glucose (pH 7.2, osmolarity of ~300 mOsm). The apical coil of the organ of Corti was dissected and placed under a grid in the recording chamber. Pillar or modiolar supporting cells were removed using soda glass pipettes in order to gain access to the basolateral face of the IHCs and to the postsynaptic boutons of type I SGNs. Dissection of the organ of Corti and cleaning of the supporting cells were performed at room temperature (20-25°C).

Electrophysiological recordings

Pre- and postsynaptic paired patch clamp recordings were performed at near physiological temperature (32-37°C) using an EPC-9 amplifier (HEKA electronics) (**Fig. 1** [↗](#)). Patch electrodes were positioned using a PatchStar micromanipulator (Scientifica, UK). Whole-cell recordings from IHCs were achieved using the perforated-patch clamp technique (Moser and Beutner, 2000 [↗](#)) using SylgardTM-coated 1.5 mm borosilicate pipettes with typical resistances between 3.5 and 6 MΩ. The IHC pipette solution contained: 129 mM Cs-gluconate, 10 mM tetraethylammonium (TEA)-Cl, 10 mM 4-AP, 10 mM HEPES, 1 mM MgCl₂ (pH 7.2, osmolarity of ~290 mOsm), as well as 300 µg/ml amphotericin B added prior to the experiment. Once the series resistance of the IHC reached below 30 MΩ, whole-cell voltage-clamp recordings from a contacting bouton was established as described in previous studies (Glowatzki and Fuchs, 2002 [↗](#); Grant et al., 2011 [↗](#); Huang and Moser, 2018 [↗](#)). For 2 pairs, the bouton recording was established first and then the IHC. SylgardTM-coated 1.0 mm borosilicate pipettes with typical resistances between 7 and 12 MΩ were used for the postsynaptic recordings. The bouton pipette solution contained: 137 mM KCl, 5 mM EGTA, 5 mM HEPES, 1 mM Na₂-GTP, 2.5 mM Na₂-ATP, 3.5 mM MgCl₂·6H₂O and 0.1 mM CaCl₂ (pH 7.2 and osmolarity of ~290 mOsm). The organ of Corti was continuously perfused with an extracellular solution containing 4.2 mM KCl, 95-100 mM NaCl, 25 mM NaHCO₃, 30 mM TEA-Cl, 1mM Na- Pyruvate, 0.7 mM NH₂PO₄·H₂O, 1mM CsCl, 1 mM MgCl₂·H₂O, 1.3 mM CaCl₂, and 11.1 mM D- glucose (pH 7.3, osmolarity of ~310 mOsm). 2.5 µM tetrodotoxin (Tocris or Santa Cruz) was added to block voltage-gated Na⁺ channels in the postsynaptic bouton.

Data were acquired using the Patchmaster software (HEKA electronics). The current signal was filtered at 5-10 kHz and sampled at 20-50 kHz. IHC were voltage-clamped at a holding potential of -58 mV, around the presumed *in vivo* resting potential (Johnson, 2015). The bouton was held at a potential of -94 mV. All reported potentials were corrected for the liquid junction potential (19 mV for the IHC and 4 mV for the bouton), measured experimentally. Ca^{2+} current recordings were corrected for the linear leak current using a P/n protocol. We excluded IHCs and boutons with leak currents exceeding -60 pA and -100 pA at holding potential, respectively. Average IHC series resistance (R_s) was $14.7 \pm 0.8 \text{ M}\Omega$ ($14.63 \pm 1.04 \text{ M}\Omega$ for high SR synapses vs $14.83 \pm 1.07 \text{ M}\Omega$ for low SR; $p = 0.7433$, Mann-Whitney U test). Average IHC membrane capacitance (C_m) was $8.8 \pm 0.19 \text{ pF}$ ($8.8 \pm 0.25 \text{ pF}$ for recordings of high SR synapses vs. $8.8 \pm 0.25 \text{ pF}$ for low SR; $p = 0.6237$, Mann-Whitney U test). The apparent series resistance of the bouton was calculated from the capacitive transient in response to a 10-mV test pulse. The actual R_s was offline calculated as reported in (Huang and Moser, 2018). Briefly, we fitted the decay phase of the capacitive transient with a double exponential. Average bouton R_s from paired recordings was $64.6 \pm 3.3 \text{ M}\Omega$ ($60.2 \pm 5.3 \text{ M}\Omega$ for high SR synapses vs $57.8 \pm 3.6 \text{ M}\Omega$ for low SR; $p = 0.7115$, unpaired t-test; and $57.01 \pm 4.7 \text{ M}\Omega$ for bouton recordings only). [Average R_s from the bouton recordings from **Figure 2-figure supplement 1** was $57 \pm 4.7 \text{ M}\Omega$]. Bouton capacitance was estimated from the area under the fast component of the double exponential fit. Average bouton C_m was $1.7 \pm 0.09 \text{ pF}$ ($1.8 \pm 0.19 \text{ pF}$ for high SR synapses vs $1.7 \pm 0.11 \text{ pF}$ for low SR; $p = 0.6575$, unpaired t-test). Average bouton membrane resistance (R_m) was $1491 \pm 133.2 \text{ M}\Omega$ ($1499 \pm 193.3 \text{ M}\Omega$ for high SR synapses vs $1487 \pm 174.3 \text{ M}\Omega$ for low SR; $p = 0.7143$, Mann-Whitney U test). The properties of the recordings (i.e., amplitude of EPSC) were not correlated with the bouton R_s (**Figure 1-figure supplement 1**).

The threshold for sEPSC detection was 4 times SD of the baseline. Spontaneous activity was calculated from time windows without stimulation with the IHC held at -58 mV; either from a 5 – 10 s recording or by averaging the number of events from the segments before and after a depolarizing pulse (**Fig. 1B**, Supplementary Table 1). To study the depletion and recovery of the pool of vesicles, we used a protocol adapted from the forward masking protocol performed during *in vivo* extracellular recordings of SGNs (Harris and Dallos, 1979; Jean et al., 2018). It consisted of two consecutive depolarizing pulses to the voltage that elicited the highest peak of Ca^{2+} current (-19 mV; **Fig. 1C**). The first pulse, called masker, lasted 100 ms and it was followed by a second pulse, called probe, which lasted 15 ms. The two pulses were separated by intervals without depolarization (interstimulus intervals, ISI) that lasted 4, 16, 64 and 256 ms. The waiting time between masker and masker was 20 s and each protocol was repeated between 3 – 20 times. To study the dynamic voltage range of synaptic transmission, we used a current-voltage (IV) protocol with 10 ms pulses of increasing voltage (from -70 mV/-60 mV to 70 mV in 5 mV steps). The interval between two stimuli was 1.5 s.

Data Analysis

Electrophysiological data was analyzed using the IgorPro 6 Software Package (Wavemetrics), GraphPad Prism 9 and Excel. Ca^{2+} charge (Q_{Ca}) and EPSC charge (Q_{EPSC}) were estimated by taking the integral of the current. Kinetics of sEPSCs, such as amplitude, 10-90% rise time, time constant of decay (τ_{decay}) and full-width half-maximum (FWHM), were calculated with Neuromatic (Rothman and Silver, 2018).

To obtain IV curves, we averaged the evoked Ca^{2+} -currents (I_{Ca}) during 5 to 10 ms after the start of each depolarization. Fractional activation of the Ca^{2+} channels was obtained from the normalized chord conductance, g ,

$$g = \frac{I}{(V - V_{\text{rev}})}$$

where V is the membrane potential and V_{rev} is the reversal potential determined by fitting a line function between the voltage of $I_{Ca\ peak} + 10\text{ mV}$ and the maximal depolarization. The activation curve was approximated by a first-order Boltzmann equation:

$$g = \frac{g_{max}}{1 + \exp\left(\frac{V_{half\ Ca} - V}{S}\right)}$$

where g_{max} is the maximum chord conductance, $V_{half\ Ca}$ is the membrane potential at which the conductance is half activated, and S is the slope factor describing the voltage sensitivity of activation.

Release intensity curves were obtained by calculating Q_{EPSC} by the end of each depolarization step and fitted using a sigmoidal function:

$$Q = \frac{Q_{max}}{1 + \exp\left(\frac{V_{h\ EPSC} - V}{rate}\right)}$$

where Q_{max} is the maximal Q_{EPSC} (normalized to 1), $V_{h\ EPSC}$ corresponds to the voltage of half-maximal release (or $Q_{50, EPSC}$) and Q is the EPSC charge. The dynamic range was determined as the voltage range between 10% and 90% of the maximal vesicle release. For statistical analysis of dynamic range, we included only pairs for which both the Ca^{2+} fractional activation and the rate level curves were possible to fit.

The apparent Ca^{2+} dependence of neurotransmitter release was studied from the 10 ms step-depolarizations of the IV curves. The resulting Q_{EPSC} vs IHC Q_{Ca} plots from each individual pair were fitted with a power function:

$$Q_{EPSC} = a(Q_{Ca})^m$$

where m corresponds to the Ca^{2+} cooperativity. Some pairs showed a clear saturation of release at high IHC Q_{Ca} . In these cases, the fit was restricted to the datapoints before the plateau, which was determined by visual inspection. For the pooled data, the power function was fitted to the normalized Q_{EPSC} vs normalized Q_{Ca} . For the pairs with saturation of release, Q_{Ca} was normalized to a point before the plateau.

For forward masking experiments, the postsynaptic response was averaged for all the repetitions for each paired recording (between 3 and 20, depending on the stability of the pair). Single active zone pool dynamics were determined by fitting an exponential plus line function to the to the first 50 ms of the average Q_{EPSC} trace in response to the masker stimulus for each ISI,

$$y_0 + A1 \left(1 - \exp\left\{\frac{-(x-x_0)}{\tau}\right\}\right) + (x - x_0)slope$$

where $A1$ is the amplitude of the exponential component, τ is the time constant of the exponential component. RRP size (in synaptic vesicles) was estimated from dividing $A1$ by the charge of the average sEPSC for each pair. Sustained exocytosis rate (in SV per s) was calculated from the slope of the linear function divided the charge of the average sEPSC. Individual recovery kinetics were determined from the ratio of probe and masker responses at 10 ms of the depolarization, with the ratio between masker and masker being 1. The recovery traces were fitted with a single exponential function from 16 ms to 20 000 ms to determine the time constant of RRP recovery.

Data was prepared for presentation using Adobe Illustrator. Skewness analysis, PCA and other statistical analysis were performed using GraphPad Prism 9. Statistical significance was assessed with unpaired t-test or non-parametric Mann-Whitney U test depending on the normal distribution and equality of variances of the data (Saphiro-Wilk test and F test). Data is expressed as mean $\pm$ sem.

Author Contributions

TM and LMJT designed the study. LMJT performed the experiments and the analysis. TM and LMJT prepared the manuscript.

Competing Interest Statement

The authors declare that no competing interests exist.

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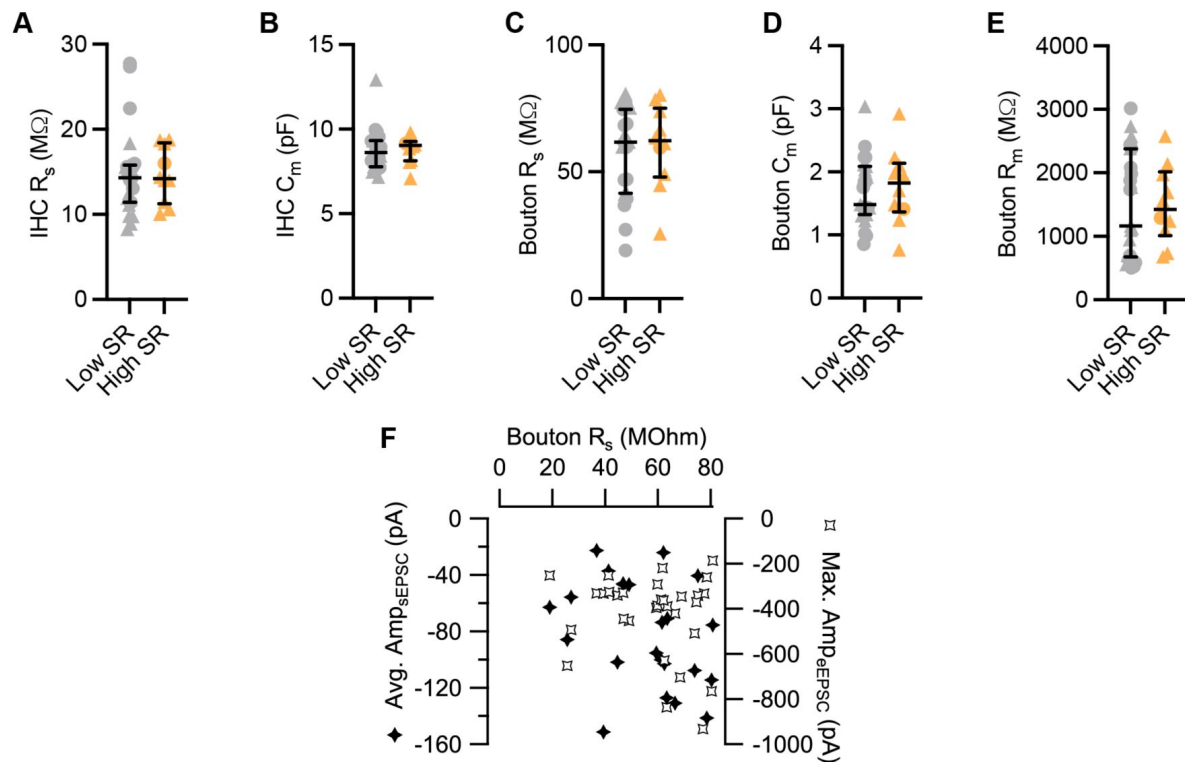


Figure 1-figure supplement 1.

Passive electrical properties of IHCs and SGN boutons

A-E. Values for IHC's series resistance (R_s , A) and membrane capacitance (C_m , B), and bouton's R_s (C), C_m (D) and membrane resistance (R_m , E) were comparable between paired recordings with low and high SR. **G.** Average sEPSC amplitudes and maximal eEPSC amplitudes plotted against bouton R_s .

Panels A-E show individual data points with the median and interquartile range overlaid (line). Synapses were classified as $\triangle$ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

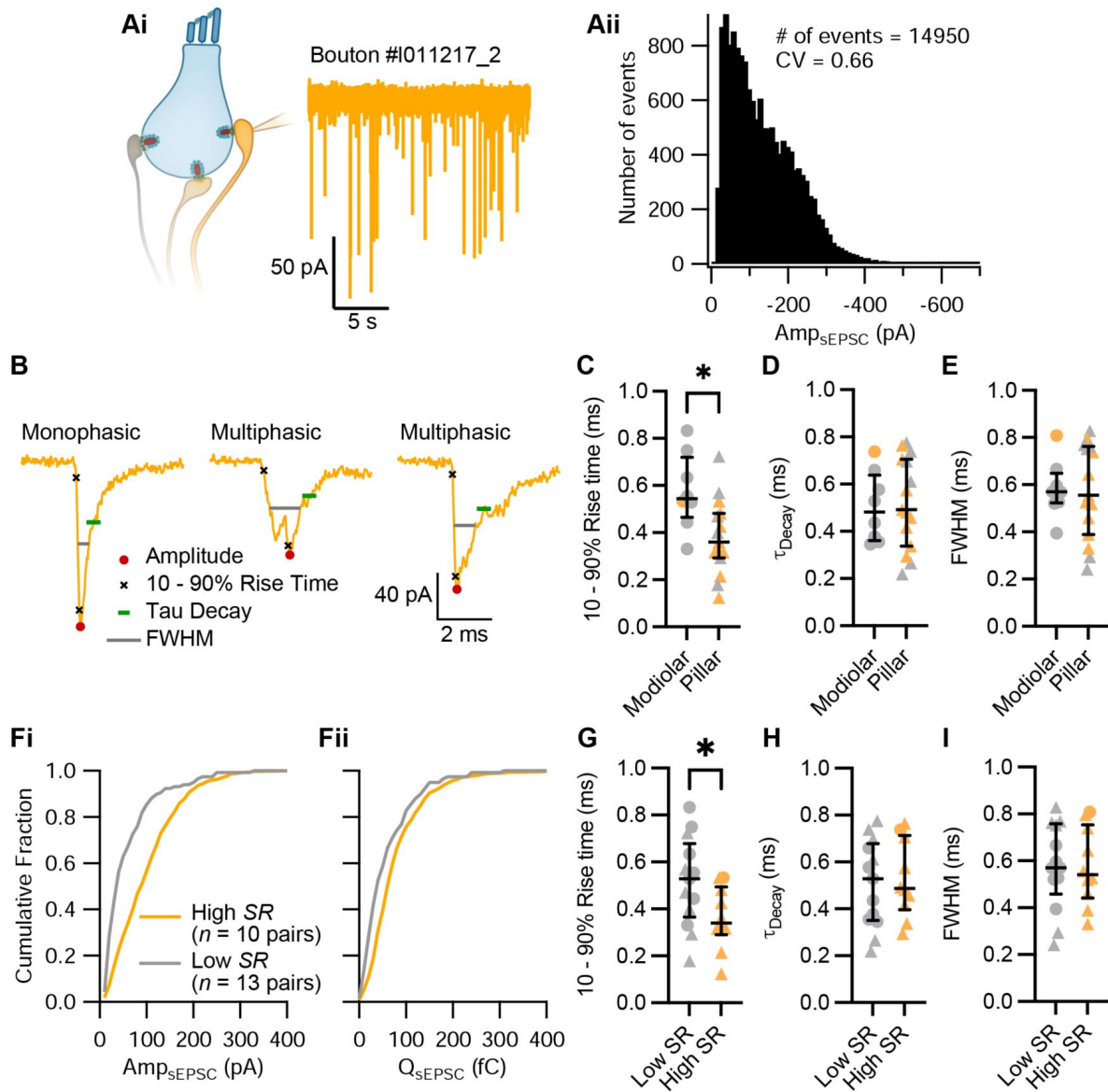


Figure 2-figure supplement 1.

Pillar and high SR synapses have sEPSCs with faster rising times.

Ai-Aii. Ruptured-patch whole-cell recordings from postsynaptic SGN boutons (without the IHC) displayed spontaneous rates from 0 up to 16.33 sEPSC/s. Pooled sEPSC amplitude (Aii) show a peak at -40 pA, with amplitudes ranging from around -10 pA to -670 pA. Bin size: 10 pA. **B.** Kinetics of sEPSCs, such as amplitude, 10-90% rise time, time constant of decay (τ_{decay}) and full-width half-maximum (FWHM), were calculated with Neuromatic (Rothman and Silver, 2018). **C-E.** Pillar synapses had faster 10-90% rise times (C) than modiolar synapses, while their τ_{decay} (D) and FWHM (E) were comparable. **Fi-Fii.** Cumulative fraction of amplitude and charge of sEPSCs from low and high SR synapses. **G-I.** High SR synapses had faster 10-90% rise times (G) than modiolar synapses, while their τ_{decay} (H) and FWHM (I) were comparable. Panels C-E, G-I show individual data points with the median and interquartile range overlaid (line). Synapses were classified as Δ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

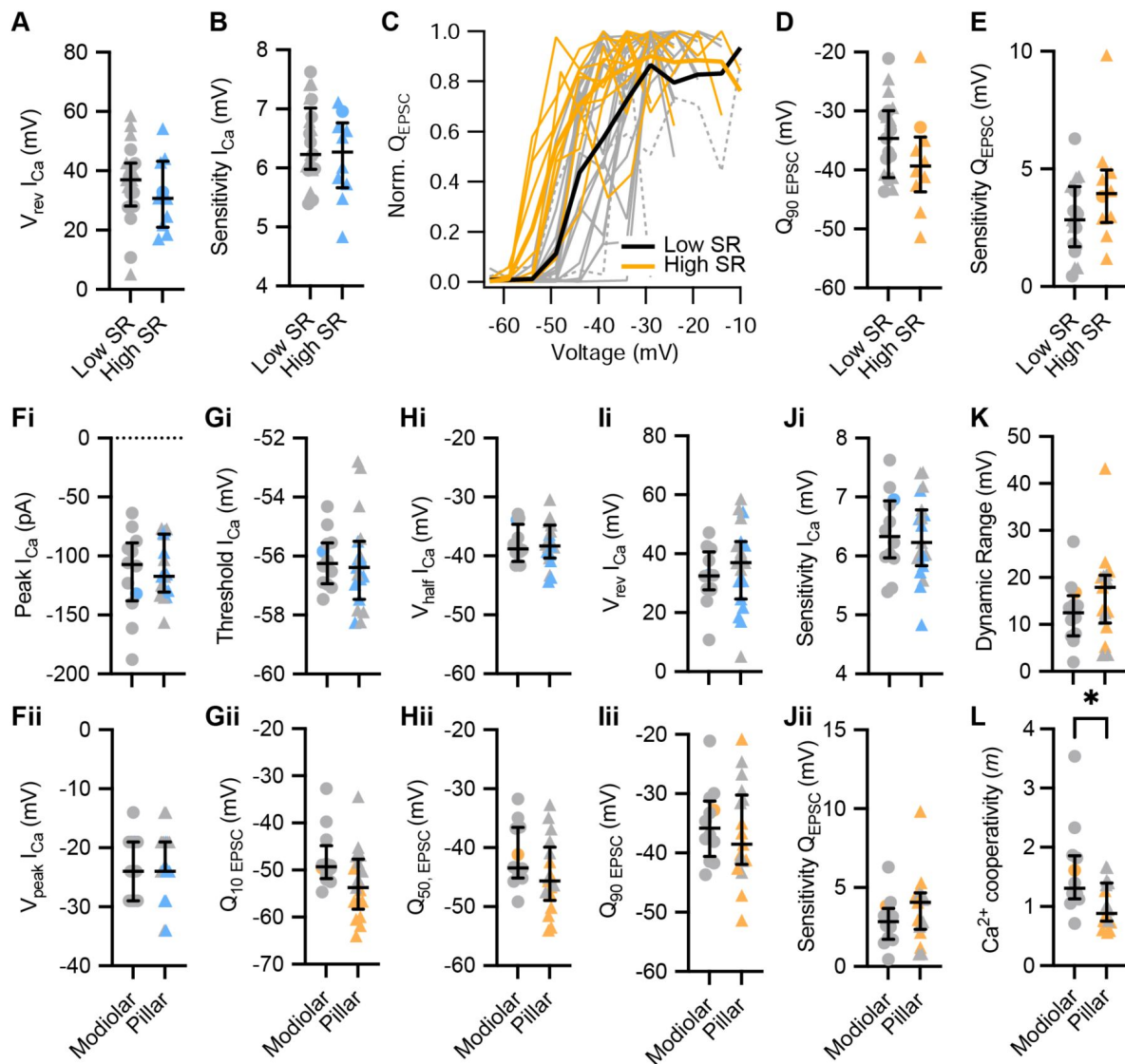


Figure 3-figure supplement 1.

The voltage dependence of synaptic release does not differ significantly between modiolar and pillar synapses, but the Ca^{2+} dependence does

A, B. The reversal potential (A) and the voltage sensitivity of I_{Ca} (B) were comparable between low and high SR synapses. **C.** Triggered single active zone eEPSCs (release intensity curves) of 31 pairs of low and high SR. Averages (thick lines) and individual curves (thin lines) are overlaid. The release-intensity curve of two low SR pairs could not be fitted with the sigmoidal function (grey dotted lines). **D-E.** $Q_{90} EPSC$ (D) and voltage sensitivity of the release (E) were comparable between high and low SR synapses. **Fi-Fii.** The peak of Ca^{2+} current (Fi) and the voltage eliciting maximum Ca^{2+} current (Fii) were comparable between modiolar and pillar synapses. **Gi-Ji.** The threshold of Ca^{2+} influx (Gi), $V_{half} I_{Ca}$ (Hi), reversal potential of I_{Ca} (Ii) and voltage sensitivity of I_{Ca} (Ji) did not differ between modiolar and pillar synapses. **Gii-Jii.** $Q_{10} EPSC$ (Gii), $V_{half} Q_{EPSC}$ (Hii), $Q_{90} EPSC$ (Iii) and voltage sensitivity of the release (Jii) were comparable between modiolar and pillar synapses. **K.** Dynamic range of release was comparable between modiolar and pillar synapse. **L.** Ca^{2+} cooperativity (m) estimated for each individual synapse was significantly lower in pillar synapses. All panels but C show individual data points with the median and interquartile range overlaid (line). Synapses were classified as Δ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

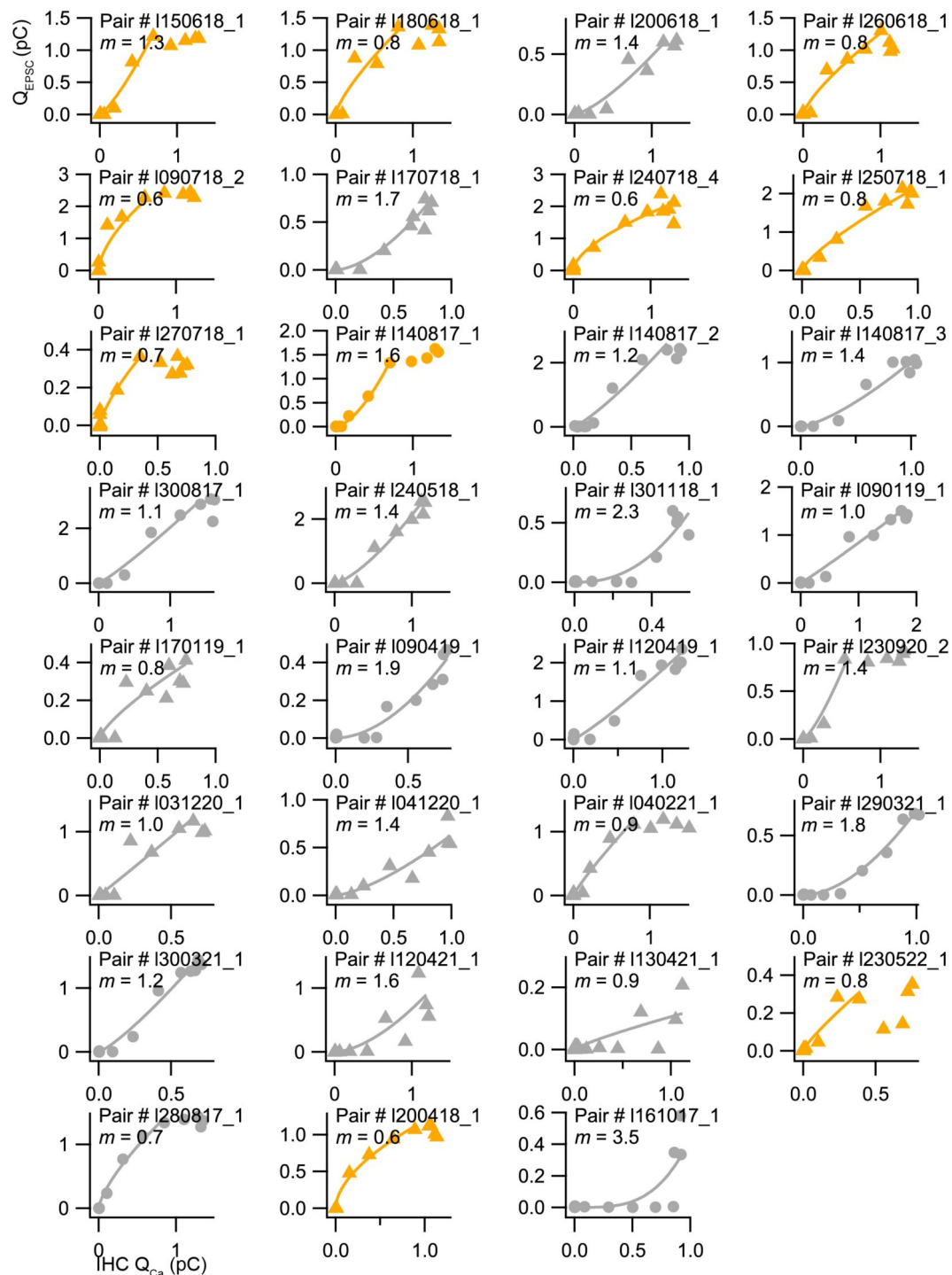


Figure 3-figure supplement 2.

Apparent Ca^{2+} dependence of neurotransmitter release at individual synapses in the range of IHC receptor potentials

Scatter plots of the EPSC charges (Q_{EPSC}) vs. the corresponding Ca^{2+} current integrals (Q_{Ca}) for each individual synapse in response to 10 ms depolarizations from -58 to -19 mV. The solid line is a least-squares fit of a power function ($Q_{EPSC} = a(Q_{Ca})^m$) to each pair data. Synapses were classified as $\triangle$ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

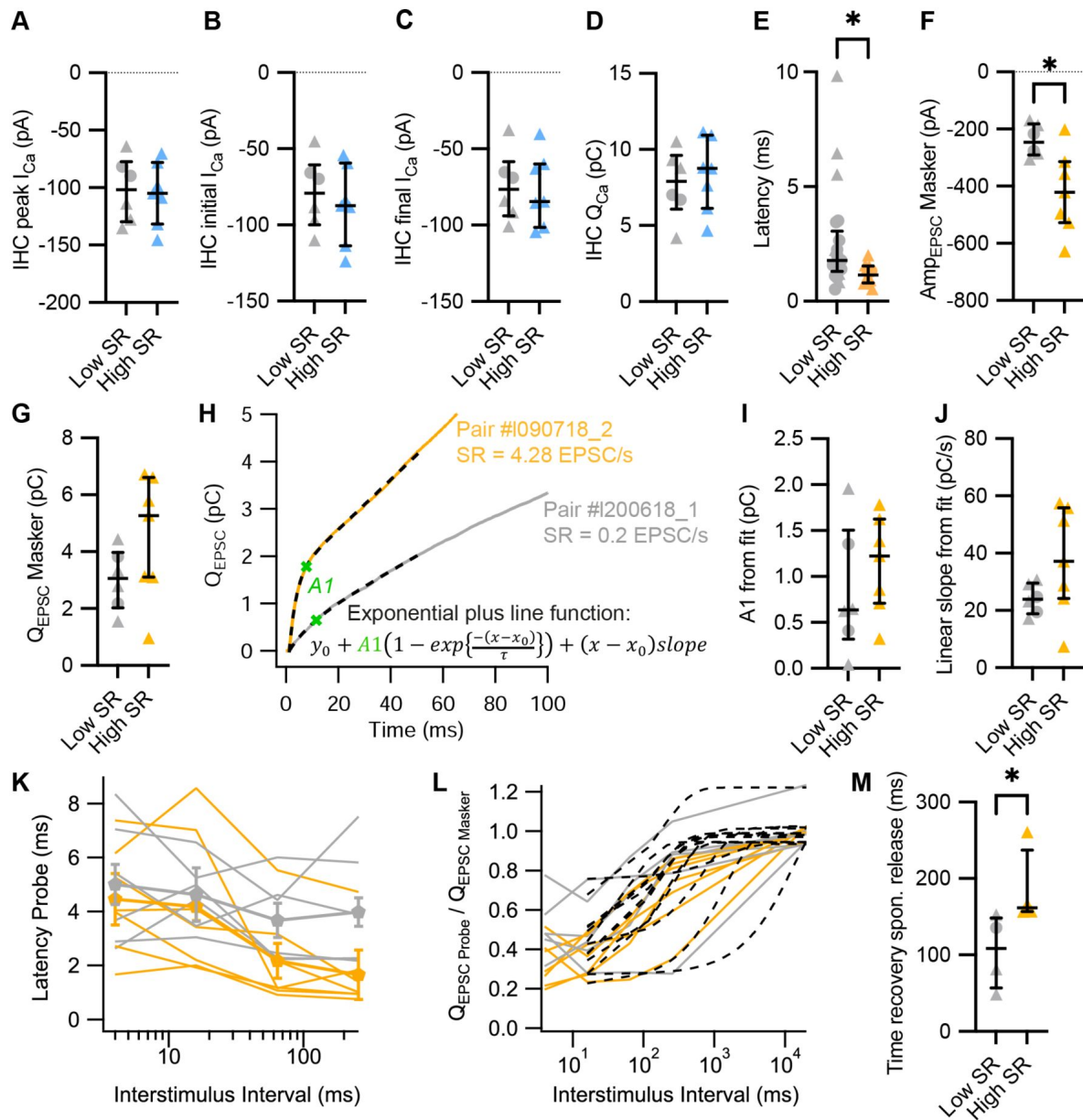


Figure 4-figure supplement 1.

Parameters of synaptic vesicle pool dynamics in forward masking protocols

A-D. Presynaptic IHC peak (A), initial (B) and final (C) Ca^{2+} current (I_{Ca}), and Ca^{2+} charge (Q_{Ca}) during the masker stimuli were comparable regardless the SR of postsynaptic bouton. **E.** Latencies of the evoked EPSCs were also significantly shorter in high SR than low SR synapses when compared in a bigger sample size (31 pairs from Fig. 4). **F-G.** High SR synapses had significantly larger amplitudes of evoked EPSCs in response to the masker stimulus (F), while the EPSC charge (Q_{EPSC}) did not differ (G). **H.** Exemplary average Q_{EPSC} response of a high SR and a low SR synapse to the masker stimuli. We fitted an exponential plus line function to the first 50 ms of the response (discontinuous lines) to study SV pool depletion dynamics. From these fits, we can retrieve information about RRP size (amplitude of the exponential component, A1), RRP depletion time constant (τ of the exponential component) and sustained release (linear slope of the line component). **I-J.** Amplitude (I) and linear slope (J) obtained from the fits of the exponential plus line function to the Q_{EPSC} from individual pairs (Fig. 4E). **K.** Latencies of the response to the probe stimuli ($EPSC_{onset} - Probe_{onset}$) for different ISI. **L.** Single exponential fits from 16 ms to 20000 ms (black dotted lines) to estimate the recovery kinetics from RRP depletion. **M.** The spontaneous release after the probe offset recovered slower in high SR synapses.

Panels A-G, I, J, M show individual data points with the median and interquartile range overlaid (line). Synapses were classified as Δ pillar or $\circ$ modiolar, and as Low SR < 1 sEPSC/s $\leq$ High SR.

Cellname	Side	SR	Total time for SR calculation	Cumulative time of 5-10 s recordings (s)	Cumulative time before/after a depolarizing pulse (s)
I140817_1	Modiolar	1.07	36		36
I140817_2	Modiolar	0.38	28.8		28.8
I140817_3	Modiolar	0.05	43.2		43.2
I300817_1	Modiolar	0.00	31.2		31.2
I240518_1	Pillar	0.22	91.2		91.2
I180618_1	Pillar	1.00	126.8		126.8
I031220_1	Pillar	0.00	10	10	
I041220_1	Pillar	0.00	10	10	
I280817_1	Modiolar	0.00	3.69		3.69
I200418_1	Pillar	3.92	3.06		3.06
I200618_1	Pillar	0.20	20.8		20.8
I260618_1	Pillar	2.05	20.8		20.8
I090718_2	Pillar	4.28	56.8		56.8
I170718_1	Pillar	0.38	23.2		23.2
I240718_4	Pillar	17.93	7.2		7.2
I250718_1	Pillar	3.18	38.4		38.4
I270718_1	Pillar	9.00	20		20
I090119_1	Modiolar	0.00	20	20	
I150618_1	Pillar	1.49	12		12
I290321_1	Modiolar	0.03	38	10	28
I300321_1	Modiolar	0.36	85.2	50	35.2
I120421_1	Pillar	0.21	35.6	10	25.6
I130421_1	Pillar	0.00	35.4	25	10.4
I301118_1	Modiolar	0.15	40	40	
I170119_1	Pillar	0.80	10	10	
I090419_1	Modiolar	0.00	10	10	
I120419_1	Modiolar	0.70	10	10	
I230920_2	Pillar	0.00	15	15	
I040221_1	Pillar	0.20	10	10	
I230522_1	Pillar	1.26	49.6	10	39.6
I230817_2	Modiolar	0.16	6.12		6.12
I161017_1	Modiolar	0.00	2.34		2.34
I200522_2	Pillar	0.00	40	40	

Supplementary Table 1.

Spontaneous activity (SR) was calculated from time windows without stimulation with the IHC held at -58 mV (Total time for SR calculation). This total time was calculated from the cumulative recording time of either from 5 – 10 s recordings and/or from the segments before and after a depolarizing pulse.

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Editors

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Reviewer #1 (Public review):

Summary:

Tobón and Moser reveal a remarkable amount of presynaptic diversity in the fundamental Ca dependent exocytosis of synaptic vesicles at the afferent fiber bouton synapse onto the pillar or mediolar sides of single inner hair cells of mice. These are landmark findings with profound implications for understanding acoustic signal encoding and presynaptic mechanisms of synaptic diversity at inner hair cell ribbon synapses. The paper will have an immediate and long-lasting impact in the field of auditory neuroscience.

Main findings: 1) Synaptic delays and jitter of masker responses are significantly shorter (synaptic delay: 1.19 ms) at high SR fibers (pillar) than at low SR fibers (mediolar; 2.57 ms). 2) Masked evoked EPSC are significantly larger in high SR than in low SR. 3) Quantal content and RRP size are 14 vesicles in both high and low SR fibers. 4) Depression is faster in high SR synapses suggesting they have a higher release probability and tighter Ca nanodomain coupling to docked vesicles. 5) Recovery of master-EPSCs from depletion is similar for high and low SR synapses, although there is a slightly faster rate for low SR synapses that have bigger synaptic ribbons, which is very interesting. 6) High SR synapses had larger and more compact (monophasic) sEPSCs, well suited to trigger rapidly and faithfully spikes. 7) High SR synapses exhibit lower voltage (~sound pressure in vivo) dependent thresholds of exocytosis.

Great care was taken to use physiological external pH buffers and physiological external Ca concentrations. Paired recordings were also performed at higher temperatures with IHCs at physiological resting membrane potentials and in more mature animals than previously done for paired recordings. This is extremely challenging because it becomes increasingly difficult to visualize bouton terminals when myelination becomes more prominent in the cochlear afferents. In addition, perforated patch recordings were used in the IHC to preserve its intracellular milieu intact and thus extend the viability of the IHCs. The experiments are tour-de-force and reveal several novel aspects of IHC ribbon synapses. The data set is rich and extensive. The analysis is detailed and compelling.

Reviewer #2 (Public review):

Summary:

The study by Jaime-Tobon & Moser is a truly major effort to bridge the gap between classical observations on how auditory neurons respond to sounds and the synaptic basis of these phenomena. The so-called spiral ganglion neurons (SGNs) are the primary auditory neurons connecting the brain with hair cells in the cochlea. They all respond to sounds increasing their firing rates, but also present multiple heterogeneities. For instance, some present a low threshold to sound intensity, whereas others have high threshold. This property inversely correlates with the spontaneous rate, i.e., the rate at which each neuron fires in the absence of any acoustic input. These characteristics, along with others, have been studied by many reports over years. However, the mechanisms that allow the hair cells-SGN synapses to drive these behaviors are not fully understood.

The level of experimental complexity described in this manuscript is unparalleled, producing data that is hardly found elsewhere. The authors provide strong proof for heterogeneity in transmitter release thresholds at individual synapses and they do so in an extremely complex experimental settings. In addition, the authors found other specific differences such as in synaptic latency and max EPSCs. A reasonable effort is put in bridging these observations with those extensively reported in in vivo SGNs recordings. Similarities are many and differences are not particularly worrying as experimental conditions cannot be perfectly matched, despite the authors' efforts in minimizing them.

<https://doi.org/10.7554/eLife.93749.2.sa2>**Reviewer #3 (Public review):**

Summary:

The manuscript by Jaime Tobon and Moser uses patch-clamp electrophysiology in cochlear preparations to probe the pre- and post-synaptic specializations that give rise to diverse activity of spiral ganglion afferent neurons (SGN). The experiments are quite an achievement! They use paired recordings from pre-synaptic cochlear inner hair cells (IHC) that allow precise control of voltage and therefore calcium influx, with post-synaptic recordings from type I SGN boutons directly opposed to the IHC for both presynaptic control of membrane voltage and post-synaptic measurement of synaptic function with great temporal resolution.

Any of these techniques by themselves are challenging, but the authors do them in pairs, at physiological temperatures, and in hearing animals, all of which combined make these experiments a real tour de force. The data is carefully analyzed and presented, and the results are convincing. In particular, the authors demonstrate that post-synaptic features that contribute to the spontaneous rate (SR) of predominantly monophasic post-synaptic currents (PSCs), shorter EPSC latency, and higher PSC rates are directly paired with pre-synaptic features such as a lower IHC voltage activation and tighter calcium channel coupling for release to give a higher probability of release and subsequent increase in synaptic depression. Importantly, IHCs paired with Low and High SR afferent fibers had the same total calcium currents, indicating that the same IHC can connect to both low and high SR fibers. These fibers also followed expected organizational patterns, with high SR fibers primarily contacting the pillar IHC face and low SR fibers primarily contacting the modiolar face. The authors also use in vivo-like stimulation paradigms to show different RRP and release

dynamics that are similar to results from SGN in vivo recordings. Overall, this work systematically examines many features giving rise to specializations and diversity of SGN neurons.

<https://doi.org/10.7554/eLife.93749.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Tobón and Moser reveal a remarkable amount of presynaptic diversity in the fundamental Ca dependent exocytosis of synaptic vesicles at the afferent fiber bouton synapse onto the pillar or mediolar sides of single inner hair cells of mice. These are landmark findings with profound implications for understanding acoustic signal encoding and presynaptic mechanisms of synaptic diversity at inner hair cell ribbon synapses. The paper will have an immediate and long-lasting impact in the field of auditory neuroscience.

Main findings: 1) Synaptic delays and jitter of masker responses are significantly shorter (synaptic delay: 1.19 ms) at high SR fibers (pillar) than at low SR fibers (mediolar; 2.57 ms). 2) Masked evoked EPSC are significantly larger in high SR than in low SR. 3) Quantal content and RRP size are 14 vesicles in both high and low SR fibers. 4) Depression is faster in high SR synapses suggesting they have a higher release probability and tighter Ca nanodomain coupling to docked vesicles. 5) Recovery of master-EPSCs from depletion is similar for high and low SR synapses, although there is a slightly faster rate for low SR synapses that have bigger synaptic ribbons, which is very interesting. 6) High SR synapses had larger and more compact (monophasic) sEPSCs, well suited to trigger rapidly and faithfully spikes. 7) High SR synapses exhibit lower voltage (~sound pressure in vivo) dependent thresholds of exocytosis.

Strengths:

Great care was taken to use physiological external pH buffers and physiological external Ca concentrations. Paired recordings were also performed at higher temperatures with IHCs at physiological resting membrane potentials and in more mature animals than previously done for paired recordings. This is extremely challenging because it becomes increasingly difficult to visualize bouton terminals when myelination becomes more prominent in the cochlear afferents.

In addition, perforated patch recordings were used in the IHC to preserve its intracellular milieu intact and thus extend the viability of the IHCs. The experiments are tour-de-force and reveal several novel aspects of IHC ribbon synapses. The data set is rich and extensive. The analysis is detailed and compelling.

We would like to thank the reviewer for the appreciation of our work and the comments that helped us to improve our manuscript. We detail our responses to the comments below.

Weaknesses:

(1) Materials and Methods: Please provide whole-cell R_s (series resistance) and C_m (membrane capacitance) average $\pm$ S.E.M. (or SD) values for IHC and afferent fiber bouton recordings. The C_m values for afferents have been estimated to be about 0.1 pF (Glowatzki and Fuchs, 2002) and it would be interesting to know if there are differences in these numbers for high and low SR afferents. Is it possible to estimate C_m from the capacitive transient time constant? Minimal electronic filtering would be required for that to work, so I realize the authors may not have this data and I also realize that the long cable of the afferents do not allow accurate C_m measurements, but some first order estimate would be very interesting to report, if possible.

In response to the reviewer's comment, we now added the estimates of series resistance and membrane capacitance for IHC and bouton recordings in Material and Methods and in the Figure 1 – figure supplement 1. Our estimate for bouton C_m is on average 1.7 ± 0.09 pF, a value that compares well to the literature. For example, Glowatzki and Fuchs (2002) provided estimates ranging 0.5-2 pF for recordings from afferent inner hair cell synapses in rats that showed a capacitance transient. In own prior work on afferent inner hair cell synapses of pre-hearing mice, we found estimates of 2.6 ± 0.5 pF (Chapochnikov et al., 2014) and 1.9 ± 0.2 pF (Takago et al., 2019). Keen and Hudspeth (2006) reported capacitances of 1–4 pF for afferent terminals in the bullfrog amphibian papilla. There was no difference in bouton C_m between high SR (1.78 ± 0.19 pF) and low SR synapses (1.68 ± 0.11 pF; $p = 0.6575$, unpaired t test).

(2) Page 20, 26 and Figure 4: With regard to synaptic delays at auditory hair cell synapses: please see extensive studies done in Figure 11 of Chen and von Gersdorff (JNeurosci., 2019); this showed that synaptic delays are 1.26 ms in adult bullfrog auditory hair cells at 31°C, which is very similar to the High SR fibers (1.19 ms; Fig.4B and page 20). During ongoing depolarizations (e.g. during a sustained sine wave) the synaptic delay can be reduced to just 0.72 ms for probe EPSCs, which is a more usual number for mature fast synapses. This paper should, thus, be cited and briefly discussed in the Discussion. So a significant shortening of delay occurs for the probe response and this is also observed in young rat IHC synapses (see Goutman and Glowatzki, 2011).

We thank the reviewer for this comment. We have analysed the synaptic delay of the probe response and included it in Figure 4 – figure supplement 1. Contrary to the findings from Goutman and Glowatzki (2011) and Chen and von Gersdorff (2019), we did not observe a shortening of the synaptic delay for the probe response compared to the masker response. This difference might arise from the duration of the masker stimulus and/or the IHC holding potential. Synaptic facilitation in hair cells seems to occur only when the RRP is not depleted by the first stimulus (Cho et al., 2011). Our 100 ms masker depolarization from a holding potential of -58 mV effectively depleted the synapse RRP (Figure 4D), while both studies mentioned above used relatively short depolarizations (2 in rat and 20 ms in bullfrog) from a holding potential around -90 mV, which most likely didn't deplete the RRP. Indeed, when using partially RRP depleting stimuli of 10 ms, Goutman (2011) observed longer synaptic latencies and smaller responses to the second stimulus. We have included this discussion in the last paragraph of the results section.

Additionally, we would also like to note that we referred to the important work on frog hair cell synapses in the manuscript, yet aimed to focus on relating synaptic heterogeneity of mammalian inner hair cell synapses to the functional diversity of type I spiral ganglion neurons that unlike the frog afferents show little branching of their peripheral neurites (in only ~15% of the neurons). We think it will be very interesting to study the aspect of presynaptic heterogeneity in the bullfrog amphibian papilla, but assume that the converging input of several active zones onto a single afferent might provide a different encoding scheme than in the mammalian cochlea.

(3) Gaussian-like (and/or multi-peak) EPSC amplitude distributions were obtained in more mature rat IHCs by Grant et al. (see their Figure 4G; JNeurosci. 2010; postnatal day 19-21). The putative single quanta peak was at 50 pA and the main peak was at 375 pA. The large mean suggests a low CV (probably < 0.4). However, Fig. 2F shows a mean of about 100 pA and CV = 0.7 for spontaneous EPSCs. This major difference deserves some more discussion. I suppose that one possible explanation may be that the current paper holds the IHC membrane potential fixed at -58 mV, whereas Grant et al. (2010) did not control the IHC membrane potential and spontaneous fluctuations in the Vm may have depolarized the IHC, thus producing larger evoked EPSCs that are triggered by Ca channel openings. Some discussion that compares these differences and possible explanations would be quite useful for the readers.

We understand the reviewer's concern. We have now included the amplitude distribution of sEPSCs recorded from 12 boutons without patch-clamping the IHC (Figure 2–figure supplement 1, panel A). The rest of the recording conditions (i.e., artificial perilymph-like solution, physiological temperature and age) were identical to the conditions used for the paired recordings. Both the range of spontaneous rate (0 up to 16.33 sEPSC/s) and the amplitude distribution (peak at -40 pA and CV of 0.66) were comparable to the values we obtained when clamping the IHC resting potential at -58 mV. In addition, for two of our pairs, we established the bouton recording first, measured the spontaneous release, then established the perforated patch-clamp of the IHC and measured the spontaneous release again with IHC held at -58 mV. For pair #1300321_1, the SR before clamping the IHC was 0.0125 sEPSC/s, with a maximal AmpsEPSC of -110 pA (avg. -52 pA). The SR while holding the IHC at -58 mV was 0.36 sEPSCs/s, with a maximal AmpsEPSC of -140 pA (avg. -46 pA). For pair #1200522_2, the SR changed from 0.07 sEPSC/s to 0. The maximal AmpsEPSC before clamping the IHC was -70 pA (avg. -31 pA). Overall, our data recorded without controlling the IHC argues against the resting potential of -58 mV as a major source of differences in EPSC rate and amplitudes compared to previous studies.

Nonetheless, it is important to note that the experimental conditions used in our study differ from previous reports in several aspects. Our extracellular solution contains the physiological pH buffer bicarbonate instead of the fast buffer HEPES, as well as TEA and Cs⁺ for proper isolation of the Ca²⁺ currents. Both pH and potassium channel blockers can alter the excitability of the cell and, consequently, the spontaneous and evoked release. For instance, despite maintaining a similar extracellular pH (7.3 to 7.4), the choice of bicarbonate or HEPES for the extracellular solution can influence differently the regulation of the intracellular pH of the cell (Michl et al., 2019). Indeed, the activity of ion channels and receptors (e.g., AMPAR), and the resting potential can change depending on the extracellular buffer used (Hare and Owen, 1998, Vincent et al., 2019, Cho and von Gersdorff, 2014; and review Sinning and Hübner, 2013). Additionally, the animal model and the age range could be a source of difference. In rats, the EPSC amplitude distribution seems to change with maturation but not with K⁺ stimulation (Grant et al., 2010) or voltage depolarizations (Goutman and Glowatzki, 2007). This however does not seem to be the case for afferent boutons recorded from mice. In resting conditions (i.e. 5.8 mM extracellular K⁺), average EPSC amplitudes are around -100 to -150 pA for both prehearing (Chapochnikov et al., 2014) and hearing mice (Niwa et al., 2021 and the present study). Upon stimulation (40 mM K⁺ or voltage depolarizations), the mean EPSC amplitude does not change in prehearing mice (Jing et al., 2013; Takaba et al., 2019), but it significantly increases in hearing mice (Niwa et al., 2021 and the present study). In p20 and p30 mice, the mean EPSC amplitude was predominantly below -100 pA at rest and only increased above -100 pA after stimulation with 40 mM K⁺ (Niwa et al., 2021). Similarly, our reported avg. AmpsEPSC is below -150 pA, while the evoked EPSCs reached average amplitudes above -200 pA (Figure 1–figure supplement 1, panel F and Figure 4 – figure supplement 1, panel F).

We have included the aforementioned points in the discussion under the section "Diversity of spontaneous release and their topographical segregation".

Reviewer #2 (Public Review):

Summary:

The study by Jaime-Tobon & Moser is a truly major effort to bridge the gap between classical observations on how auditory neurons respond to sounds and the synaptic basis of these phenomena. The so-called spiral ganglion neurons (SGNs) are the primary auditory neurons connecting the brain with hair cells in the cochlea. They all respond to sounds increasing their firing rates, but also present multiple heterogeneities. For instance, some present a low threshold to sound intensity, whereas others have high threshold. This property inversely correlates with the spontaneous rate, i.e., the rate at which each neuron fires in the absence of any acoustic input. These characteristics, along with others, have been studied by many reports over the years. However, the mechanisms that allow the hair cells-SGN synapses to drive these behaviors are not fully understood.

Strengths:

The level of experimental complexity described in this manuscript is unparalleled, producing data that is hardly found elsewhere. The authors provide strong proof for heterogeneity in transmitter release thresholds at individual synapses and they do so in extremely complex experimental settings. In addition, the authors found other specific differences such as in synaptic latency and max EPSCs. A reasonable effort is put into bridging these observations with those extensively reported in in vivo SGNs recordings. Similarities are many and differences are not particularly worrying as experimental conditions cannot be perfectly matched, despite the authors' efforts in minimizing them.

We would like to thank the reviewer for the appreciation of our work and the comments that helped us to improve our manuscript. We detail our responses to the comments below.

Weaknesses:

Some concern surges in relation to mismatches with previous reports of IHC-SGN synapses function. EPSCs at these synapses present a peculiar distribution of amplitudes, shapes, and rates. These characteristics are well-established and some do not seem to be paralleled in this study. Here, amplitude distributions are drastically shifted to smaller values, and rates of events are very low, all compared with previous evidence. The reasons for these discrepancies are unclear. The rate at which spontaneous EPSCs appear is an especially sensitive matter. A great part of the conclusions relies on the definition of which of the SGNs (or should say synapses) belong to the low end and which to the high end in the spectrum of spontaneous rates. The data presented by the authors seem a bit off and the criteria used to classify recordings are not well justified. The authors should clarify the origin of these differences since they do not seem to come from obvious reasons such as animal ages, recording techniques, mouse strain, or even species.

We understand the reviewer's concern. We have now included the amplitude distribution of sEPSCs recorded from 12 boutons without patch-clamping the IHC (Figure 2-figure supplement 1, panel A). The rest of the recording conditions (i.e., artificial perilymph-like solution, physiological temperature and age) were identical to the conditions used for the paired recordings. Both the range of spontaneous rate (0 up to 16.33 sEPSC/s) and the amplitude distribution (peak at -40 pA and CV of 0.66) were comparable to the values we

obtained when clamping the IHC resting potential at -58 mV. In addition, for two of our pairs, we established the bouton recording first, measured the spontaneous release, then established the perforated patch-clamp of the IHC and measured the spontaneous release again with IHC held at -58 mV. For pair #1300321_1, the SR before clamping the IHC was 0.0125 sEPSC/s, with a maximal AmpsEPSC of -110 pA (avg. -52 pA). The SR while holding the IHC at -58 mV was 0.36 sEPSCs/s, with a maximal AmpsEPSC of -140 pA (avg. -46 pA). For pair #1200522_2, the SR changed from 0.07 sEPSC/s to 0. The maximal AmpsEPSC before clamping the IHC was -70 pA (avg. -31 pA). Overall, our data recorded without controlling the IHC argues against the resting potential of -58 mV as a major source of differences in EPSC rate and amplitudes compared to previous studies.

Additionally, as noted on the section “Diversity of spontaneous release and their topographical segregation”, our SR values also agree with the range of 0.1 – 16.42 spikes/s reported by Wu et al., (2016) using loose patch recordings from p15-p17 rats. 90% of the paired recordings (and 60% of the bouton recordings) of our dataset were obtained from mice between p14-p17, where spontaneous activity is still low compared to older age groups (p19-p21: 0 – 44.22 spikes/s; p29p32: 0.11 – 54.9 spikes/s Wu et al., 2016; p28: 0 – 47.94 spikes/s, Siebald et al., 2023). There are two additional aspects to consider: i) about 40% of the SGN spikes seem to be generated intrinsically (not activated by an EPSP, ergo an EPSC) at p15-p18 (Wu et al., 2016); and ii) the presence of a spike or EPSC is the sole determinant of a successful recording when the IHC is not stimulated (either by K⁺ or voltage), thus, these type of experiments undersample fibers with low SR.

We have included the aforementioned points in the discussion under the section “Diversity of spontaneous release and their topographical segregation”.

Reviewer #3 (Public Review):

Summary:

"Bridging the gap between presynaptic hair cell function and neural sound encoding" by Jaime Tobon and Moser uses patch-clamp electrophysiology in cochlear preparations to probe the pre- and post-synaptic specializations that give rise to the diverse activity of spiral ganglion afferent neurons (SGN). The experiments are quite an achievement! They use paired recordings from pre-synaptic cochlear inner hair cells (IHC) that allow precise control of voltage and therefore calcium influx, with post-synaptic recordings from type I SGN boutons directly opposed to the IHC for both presynaptic control of membrane voltage and post-synaptic measurement of synaptic function with great temporal resolution.

Strengths

Any of these techniques by themselves are challenging, but the authors do them in pairs, at physiological temperatures, and in hearing animals, all of which combined make these experiments a real tour de force. The data is carefully analyzed and presented, and the results are convincing. In particular, the authors demonstrate that post-synaptic features that contribute to the spontaneous rate (SR) of predominantly monophasic post-synaptic currents (PSCs), shorter EPSC latency, and higher PSC rates are directly paired with pre-synaptic features such as a lower IHC voltage activation and tighter calcium channel coupling for release to give a higher probability of release and subsequent increase in synaptic depression. Importantly, IHCs paired with Low and High SR afferent fibers had the same total calcium currents, indicating that the same IHC can connect to both low and high SR fibers. These fibers also followed expected organizational patterns, with high SR fibers primarily contacting the pillar IHC face and low SR fibers primarily contacting the modiolar face. The authors also use in vivo-like stimulation paradigms to show different RRP and release dynamics that are similar to results from SGN in vivo

recordings. Overall, this work systematically examines many features giving rise to specializations and diversity of SGN neurons.

We would like to thank the reviewer for the appreciation of our work and the comments that helped us to improve our manuscript. We detail our responses to the comments below.

Weaknesses / Comments / edits:

(1) The careful analysis of calcium coupling and EPSC metrics is especially nice. Can the authors speculate as to why different synapses (likely in the same IHC) would have different calcium cooperativity?

The finding of different apparent Ca²⁺ cooperativities among IHC synapses is intriguing. Paired pre- and postsynaptic patch-clamp recordings (this work and (Jaime Tobón and Moser, 2023)) and single synapse imaging of presynaptic Ca²⁺ signals and glutamate release (Özçete and Moser, 2021) jointly support this notion. Both methodologies complement each other. Imaging allows to assess the presynaptic Ca²⁺ of the specific synapse, while in paired recordings release is related to the whole cell Ca²⁺ influx. Paired recordings, on the other hand, provide the sensitivity and temporal resolution to assess the initial release rate with short stimuli (2 to 10 ms), which avoids an impact of RRP depletion and ongoing SV replenishment that needs to be considered for the longer stimuli used in imaging (50 ms). Both approaches agree on the finding of tighter coupling of Ca²⁺ channels and release sites (i.e., lower apparent Ca²⁺ cooperativity during depolarization within the range of receptor potentials) at pillar synapses. Moreover, the present study took advantage of recording individual release events [which was not achieved by imaging] and further supported the hypothesis that high SR SGNs receive input from active zones with tighter coupling than low SR SGNs. However, our two non-overlapping data sets for paired patch-clamp recordings (this work and (Jaime Tobón and Moser, 2023)) found a narrower range of apparent Ca²⁺ cooperativities compared to results from single synapse imaging (Özçete and Moser, 2021). This might reflect the technical differences described above. Future studies, potentially combining paired patch-clamp recordings with imaging of presynaptic Ca²⁺ signals will be needed to scrutinize this aspect.

We think that the different Ca²⁺ cooperativities reflect subtle differences in the topography of presynaptic Ca²⁺ channels and vesicular release sites at the specific IHC active zones. The work of Özçete and Moser (2021) indicated that indeed, apparent Ca²⁺ cooperativities differ among active zones even within the same inner hair cell. Synaptic heterogeneity within one individual cell can expand its coding capacity. In the case of IHCs, differences in the Ca²⁺ dependence of synaptic release, in addition to the heterogeneous voltage dependence, appears to diversify the response properties (i.e., synaptic vesicle release probability) of individual synapses to the same stimulus. This is particularly important for sound intensity and temporal coding.

We have included the aforementioned points in the discussion under the section "Candidate mechanisms distinguishing evoked release at low and high SR synapses".

(2) On the bottom of page 6 it would be helpful to mention earlier how many pillar vs modiolar fibers were recorded from, otherwise the skewness of SRs (figure 2H could be thought to be due to predominantly recordings from modiolar fibers. As is, it reads a bit like a cliff-hanger.

Done!

(3) The contrasts for some of the data could be used to point out that while significant differences occur between low and high SR fibers, some of these differences are no

longer apparent when comparing modiolar vs pillar fibers (eg by contrasting Figure 2C and 2K). This can indicate that indeed there are differences between the fiber activity, but that the activity likely exists in a gradient across the hair cell faces. Pointing this out at the top of page 10 (end of the first paragraph) would be helpful, it would make the seemingly contradictory voltage dependence data easier to understand on first read (voltage-dependence of release is significantly different between different SR fibers (figure 3) but is not significantly different between fibers on different HC faces (figure S3).

Done!

(4) It should be acknowledged that although the use of post-hearing animals here (P14-23) ensures that SGN have begun to develop more mature activity patterns (Grant et al 2010), the features of the synapses and SGN activity may not be completely mature (Wu et al 2016 PMID: 27733610). Could this explain some of the 'challenges' (authors' section title) detailed on page 28, first full paragraph?

Done!

(5) In the discussion on page 24, the authors compare their recorded SR of EPSCs to measure values in vivo which are higher. Could this indicate that in vivo, the resting membrane potential of IHCs is more depolarized than is currently used for in vitro cochlear experiments?

That is indeed one possible explanation among others. We have expanded the discussion about the factors that could affect the SR in *ex vivo* experiments.

(6) The results showing lower calcium cooperativity of high SR fibers are powerful, but do the authors have an explanation for why the calcium cooperativity of < 2 is different from that ($m = 3-4$) observed in other manuscripts?

We assume this question to potentially result from a misunderstanding. Using membrane capacitance measurements and Ca^{2+} uncaging, Beutner et al. (2001) reported a high intrinsic Ca^{2+} cooperativity of inner hair cell exocytosis ($m = 4-5$). Based on this data, it has been proposed that the binding of 4-5 Ca^{2+} ions is required to trigger the fusion of a synaptic vesicle in IHCs. However, given the shortcoming of Ca^{2+} uncaging, we and others aimed to further study this aspect using alternative methods. By varying the current of single Ca^{2+} channels in apical IHCs of hearing mice, several studies reported a high apparent Ca^{2+} cooperativity ($m = 3-5$) that is thought to reflect the high intrinsic cooperativity (Brandt et al., 2005; Wong et al., 2014; Özçete and Moser, 2021; Jaime Tobón and Moser, 2023).

On the other hand, the apparent Ca^{2+} cooperativity observed upon changing the number of open Ca^{2+} channels would also reflect the active zone topography (i.e., number and distance of Ca^{2+} channels to the vesicular release site). In the present study, we used different depolarizations within the range of receptor potentials and found a low apparent Ca^{2+} cooperativity ($m < 2$) in 93% of the studied synapses. Other studies in apical IHCs from hearing mice used similar and alternative methods to change the number of open Ca^{2+} channels and also estimated an apparent cooperativity of < 2 (Brandt et al., 2005; Johnson et al., 2005; Johnson et al., 2007; Wong et al., 2014; Özçete and Moser, 2021; Jaime Tobón and Moser, 2023). The fact that these estimates are smaller than those seen upon changes in single Ca^{2+} current has been taken to indicate that SV release is governed by one or few Ca^{2+} channels in nanometer proximity (Ca^{2+} nanodomain-like control of SV exocytosis), building on classical synapse work (Augustine et al., 1991).

In contrast, comparable recordings from mouse IHCs before the onset of hearing (Wong et al., 2014) revealed more similar apparent Ca^{2+} cooperativities ($m \sim 3$) for both changes in the

number of open Ca^{2+} channels and changes in single Ca^{2+} channel current. This suggests that IHCs before the onset of hearing employ a Ca^{2+} microdomain-like control of SV exocytosis in which release is governed by the combined activity of several Ca^{2+} channels in >100 nm distance to the release site. A Ca^{2+} microdomain-like control of SV exocytosis was also reported for basocochlear IHCs (Johnson et al., 2017).

Recommendations for the authors:

As explained in the public reviews of Reviewers 1 and 2, some mismatches between the data presented here and previous reports from the literature have been identified. It is recommended that you discuss those mismatches, perhaps in relation to the choice of patchclamping the hair cells at -58mV.

We have addressed this point thoroughly in the revised MS. Please see our response to the public review.

Reviewer #1 (Recommendations For The Authors):

Minor suggestions and corrections:

(1) Figures 3 and 4 show beautiful data with paired recordings. Figure 3 shows 10 ms pulses, whereas Fig. 4 shows 100 ms depolarizing pulses. The example in Fig. 3A shows asynchronous release after Ca channel closure, whereas Fig. 4 does not show this so prominently. Was there quite a bit of variability in the asynchronous release from different cell pairs, or was this correlated with pulse duration?

The asynchronous release is also present after 100 ms depolarizing pulses (please see the updated panel A of Figure 4). However, we have not analysed asynchronous release and think that this would be beyond the scope of the current MS. For clarity, we have added dashed lines in the EPSC traces of Figs. 3 and 4 to indicate the on and off-set of the depolarization.

(2) Differences in apex and basal IHC ribbon synapse nanodomain to microdomain Ca channel coupling to exocytosis-sensor have been reported also for gerbil IHCs (see Johnson et al., JNeurosci., 2017). This may be worth mentioning since it is another indication of major synaptic diversity in the mammalian cochlea, this time from low to frequency-located IHCs.

Done

(3) Page 22: change "hight SR" to "high SR".

Done

(4) Page 27: change "address" to "addressed".

Done

Reviewer #2 (Recommendations For The Authors):

Major points:

(1) As indicated in methods, recording stretches of 5-10 seconds were used to determine the SR of a given SGN. This seems too short for a reasonable estimate of the SR in these neurons. Also, the reported SRs for these mature mice are not only much lower than those measured in in-vivo SGN extracellular recordings but also compared to those

reported in ex-vivo rat recordings. Why this discrepancy? The authors decided to estimate SR by voltage-clamping IHCs at a fixed value of -58 mV, which they take from Johnson, 2015. I wonder if it is not more reasonable to use a range of IHC holdings and measure SR at those, instead of using a single one. It is hard to visualize a very strong argument for using strictly -58 mV. In addition, mapping out a range of holding potentials could provide additional information on IHCs resting membrane potential in physiological conditions.

Related to this point, considering that SR values found in the ex-vivo preparation are much lower than those described in in-vivo situations, is it fair to use the same 1 sp/s criteria, as in Taberner & Liberman, to segregate low and high? Shouldn't this value be adjusted to the overall lower SR? This criterion is naturally critical for the consequent evaluation of other SGN properties.

Finally, on this same problem of IHC V_h , does -58 mV estimate include the 19 mV liquid junction potential? How does it compare with the activation threshold of calcium influx at modiolar vs pillar synapses (see imaging studies)?

We had proactively discussed the challenges of relating *ex vivo* and *in vivo* data in the preprint provided for review. While we consider the outcome of our study helpful for better understanding the relation of afferent synaptic heterogeneity and diverse firing properties of SGNs, we do not claim that the assumptions based on literature (such as on the physiological resting potential) represent ground truth.

When carefully revising the MS, we have expanded on the discussion to address the points raised here, particularly regarding the lower SR and sEPSC amplitudes. As this and the other reviewer commented in the public review, these experiments were hard to achieve and we consider repeating them with a range of IHC holding potentials (then not only for spontaneous rate of transmission, but also for in depth characterization of evoked release) to be beyond the scope of the present study.

We do appreciate the suggestion to adjust the distinction between low and high SR given the overall lower rates. However, we would like to refrain from it, as i) we consider it quite arbitrary to define another criterium and ii) we would like to avoid any apparent cherry-picking bias.

Finally, yes, of course, the -58 mV represent the liquid junction potential corrected holding potential. Our average IHC whole-cell V_{half} ICa (-38.86 mV for high SR and -37.60 mV for low SR) compares well with previous reports of average whole-cell V_{half} ICa (-35.44 mV) and average synaptic V_{half} Rhod-FF (-41.15 mV) (Özçete and Moser, 2021). Additionally, our V_{half} QEPSC distribution (ranging from -53.97 to -31.72 mV) also compares well with the V_{half} iGluSnFR distribution (ranging from -45.25 to -29.86 mV) obtained by imaging of synaptic glutamate release (Özçete and Moser, 2021).

1. EPSCs amplitude distributions in Figure 2 seem very different from those reported before by Grant et al., 2010 and Niwa et al., 2021 (even Chapochnikov et al., 2014; although not sure if the animal ages match). The average amplitudes of EPSCs reported here, for either pillar or modiolar SGNs, seem way smaller than those reported previously. The authors should provide a convincing explanation for this critical deviation from the consensual results.

Please refer to our response to the public review (point #3).

1. Rise time analysis in Fig. 2 supp 1. The actual values seem too long, again, compared to reported values. Also, what would these differences between modiolar and pillar represent?

Previous reports on mouse, rat, turtle and bullfrog focused mainly on the rise times (or time to peak) of monophasic EPSCs: about 0.39 ms (p8-p11 mouse; Chapochnikov et al., 2014, Takago et al., 2019), 0.33-0.58 ms (p7-p14 rat; Yi et al., 2010, Grant et al., 2010, Glowatzki and Fuchs, 2002), 0.17-0.29 ms (p15-p21 rat; Chapochnikov et al., 2014, Huang and Moser, 2018, Grant et al., 2010), 0.1-0.2 ms (turtle auditory papilla; Schnee et al., 2013) and 0.15-0.2 ms (bullfrog 31°C and 22°C; Li et al., 2009, Chen and von Gersdorff, 2019). Regarding multiphasic EPSCs, some studies have reported rise times (or times to peak) of about 1.5 ms (p8-p11 mouse; Takago et al., 2019), 1.1 ms (p8-p11 rat; Grant et al., 2010) and 0.6-0.8 ms (p15-p21 rats; Huang and Moser, 2018, Chapochnikov et al., 2014, Grant et al., 2010). When we factor in the waveform of the sEPSCs, our rise times are comparable to the literature:

Author response table 1.

	Synapse classification	Mean $\pm$ sem (ms)	Median (ms)	p-value
Monophasic EPSCs	Pillar	0.14 $\pm$ 0.002	0.13	$p < 0.0001$
	Modiolar	0.19 $\pm$ 0.01	0.16	Mann-Whitney U test
	High SR	0.15 $\pm$ 0.002	0.13	$p = 0.0014$
	Low SR	0.19 $\pm$ 0.01	0.18	Mann-Whitney U test
Mul-phasic EPSCs	Pillar	0.61 $\pm$ 0.03	0.45	$p = 0.0289$
	Modiolar	0.74 $\pm$ 0.08	0.56	Mann-Whitney U test
	High SR	0.62 $\pm$ 0.03	0.45	$p = 0.0422$
	Low SR	0.66 $\pm$ 0.05	0.58	Mann-Whitney U test

Thus, IHC synapses with higher SR and predominantly located at the pillar side appear to have sEPSCs with faster rise times regardless of their waveform. This might be a consequence of the fusion kinetics of the synaptic vesicles which are tightly influenced by the Ca²⁺ influx (Huang and Moser, 2018). Additionally, differences in the composition and density of the postsynaptic AMPA receptors could play a role in the rise time of the EPSC (Rubio et al., 2017).

1. One of the most impressive observations of the in-vivo SGN physiology is the difference in sound threshold among specific fibers. This can vary over tens of dB of sound pressure levels.

The representation of this phenomenon when using an ex-vivo preparation is not obvious. Overall, it has been reported that IHC Vm is a good proxy for stimulus intensity. Consequently, the authors reported an 'IHC Vm threshold' at the start of SGN synaptic activity for each recording. This can be found in Figure 3 Eii, where values vary between -65 to -30 mV. This is already an important finding. However, the representative traces on panel A only diverge by 5 mV. It would be very interesting to the reader to have represented in the figure recordings that can better illustrate this wide range of values.

We agree with the reviewer regarding the impressive difference in the sound thresholds recorded in vivo. To illustrate better our findings, we have chosen a different representative trace for the high SR synapse.

1. On the masker-probe experiments it would be interesting to look at the synaptic delay of the probe pulses. Are they different between high and low SR synapses?

We have now included the results of the synaptic delay of the probe response (Figure 4–supplementary figure 1). Despite not being statistically significant, the eEPSC probe latency of high SR is on average faster than low SR.

Reviewer #3 (Recommendations For The Authors):

(1) The terms monophasic and compact are used interchangeably. This is fine, but perhaps compact could be defined earlier, otherwise, readers may think that 'compact' means 'short' (as is sometimes euphemistically used to describe short people), which then makes phrasing such as the figure legend for figure 2 a bit confusing. This could be included at first use in a figure as well, in figure 1B where the two types of EPSCs are first shown.

Done, now explained and preferentially used monophasic.

(2) Check for mention of figure panels in the results text - for example, there is no mention in the results text of figure 2A, 2I,

Done

(3) The locations of some of the statistics are inconsistent. This is fine if the authors have a reason for including the stats where they did, but in some cases, the stats are duplicated (for example figure 2J, 2K, 2L, the stats are in both the figure legend and the results text, then check throughout).

Done

(4) The color coding in figure 4 is confusing in panel A - does orange still mean a high SR fiber here? The legend indicates that orange is for EPSCs, but does not specify charge. It could be helpful to show both a high and low SR response, both for EPSCs and for charge.

Thanks for pointing us to this aspect: we have carefully revised the figure and figure legend for clarity. We also included an exemplary response of a low SR synapse in the figure.

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